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Volumetric Comparisons of Brain Structures in Bats[1]

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With 19 Figures

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Introduction

From the point of view of their mode of locomotion, the bats (order Chiroptera) constitute a strikingly distinct group among mammals. However, in spite of the unity which they derive from the unique possession of flight-ability, bats show, in their encephalization, remarkable differences (Pirlot 1969, 1970, Pirlot and Stephan 1970). In the last paper we compared the brain sizes of 51 species from 10 families using the allometry formula. We found low encephalization in the Emballonuridae, Molossidae, Vespertilionidae, Rhinolophidae and Hipposideridae as well as in *Noctilio labialis* and several Phyllostomatidae. A median encephalization existed in the bulk of the Phyllostomatidae and a high one in *Noctilio leporinus*, the Desmodontidae and the Pteropidae. These results confirmed a fact well known from other widely ramified groups of mammals (e. g. Bauchot and Stephan 1969, in Primates), namely that encephalization can be similar within different families and quite divergent within one family. It is therefore no useful criterion in a discussion of systematic relationships. However, another and surprising relationship exists between the degree of encephalization and the feeding habits of the various species. All insect-eaters (represented by 31 species) possessed a low encephalization, and in ascending order of the mean values follow the nectar-feeders (2 species), the frugivorous Microchiroptera (7 species), the fish-eaters (1 species), the vampires (2 species) and the frugivorous Megachiroptera (8 species).

In these studies of encephalization, however, we considered only the size of the *total* brain, that is to say, the sum of encephalic structures. But as, even in cases of similar encephalization, the general organisation of the brain may vary distinctly, we then postponed the discussion of certain problems until an analysis of a variety of parts of the brain could be achieved.

The type of quantitative investigation here presented is extremely time-consuming. We therefore restricted ourselves to a limited number of species and specimens, but endeavoured to include representative forms of all Chiroptera-types that we recog-

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nized as distinct in our earlier study of encephalization. Further studies are in preparation and shall be carried out in two directions (1) in a more detailed investigation into finer brain structures (several of the parts measured here are very complex) and (2) in an investigation of carefully selected additional species and specimens.

The first results are presented here in the expectation that they already permit a general characterization of the different types on hand. They will no doubt have to be modified to some extent as more material is investigated. However, the results of future research are not expected to influence the *fundamental* conclusions reached in the present instance. Since the present results confirm the close relationship between characteristics of the brain and feeding habits in bats, an attempt was made to interpret the possible reasons of this interdependence and possible ways of its phylogenetic development.

As previously mentioned in our paper on encephalization (1970), the Chiroptera are thought to originate from insectivore-like ancestors. To facilitate a phylogenetic interpretation of our data, we have compared them with those of the most primitive recent representatives of the order Insectivora calling them "Basal Insectivores" (more detailed characterization of this group is given in STEPHAN 1967b).

We know, that all discussion of phylogenetic development based on the comparison of *recent* material is indirectly and essentially hypothetical. However, recent material can contribute well founded arguments to the problems of phylogeny. To make critical evaluation possible and to promote further analysis with other mathematical methods, all volumes so far measured are published herewith.

Material and Methods

This investigation was conducted on 20 brains from 18 species belonging to 8 families, as follows:

Noctilionidae:	*Noctilio leporinus*
Rhinolophidae:	*Rhinolophus hipposideros*
Hipposideridae:	*Hipposideros bicolor, Asellia tridens*
Phyllostomatidae:	*Phyllostomus discolor, Glossophaga soricina, Carollia perspicillata, Sturnira lilium, Vampyrops helleri, Artibeus jamaicensis, A. lituratus*
Desmodontidae:	*Desmodus rotundus*
Vespertilionidae:	*Myotis nattereri, M. myotis*
Molossidae:	*Chaerophon leucostigma*
Pteropidae:	*Cynopterus brachyotis, C. horsfieldii, Eonycteris spelaea*

With regard to the origin, the fixation and the weighing of the specimens, we refer to PIRLOT and STEPHAN (1970). In the case of *Asellia*, a difficulty arose from the fact that no good weighing scale was available at the time of collection. The weights may have been somewhat overestimated with the equipment on hand, leading to a parallel underestimation of the development of encephalic structures. However, the collector (PIRLOT) does not think that the differences could be so great as to bias our general conclusions.

The brains were embedded in paraffin and sectioned serially. The 10 µ sections were alternately stained with cresyl-violet for nerve-cells, and by the Heidenhain-Woelcke technique for myelinated fibers.

Methods for estimating and comparing the size of the brains and its parts have been described in previous publications (STEPHAN 1967a, b, PIRLOT and STEPHAN 1970). To facilitate the understanding of the quantitative results, we would like, however, briefly to repeat here the main points of those methods: 1. Volume estimation from enlarged photographs of serial sections at equal intervals (figs. 1–2). – 2. Comparison of the absolute values with the aid of the allometry formula, i. e. taking into account the differences in body size. – 3. Reference to the size of the corresponding structures in the "basal Insectivores", which are the most primitive living placentals according to characteristics of the brain (figs. 3–5). – 4. Introduction of "Progression indices" which allow a direct numerical estimate of how many times a given brain

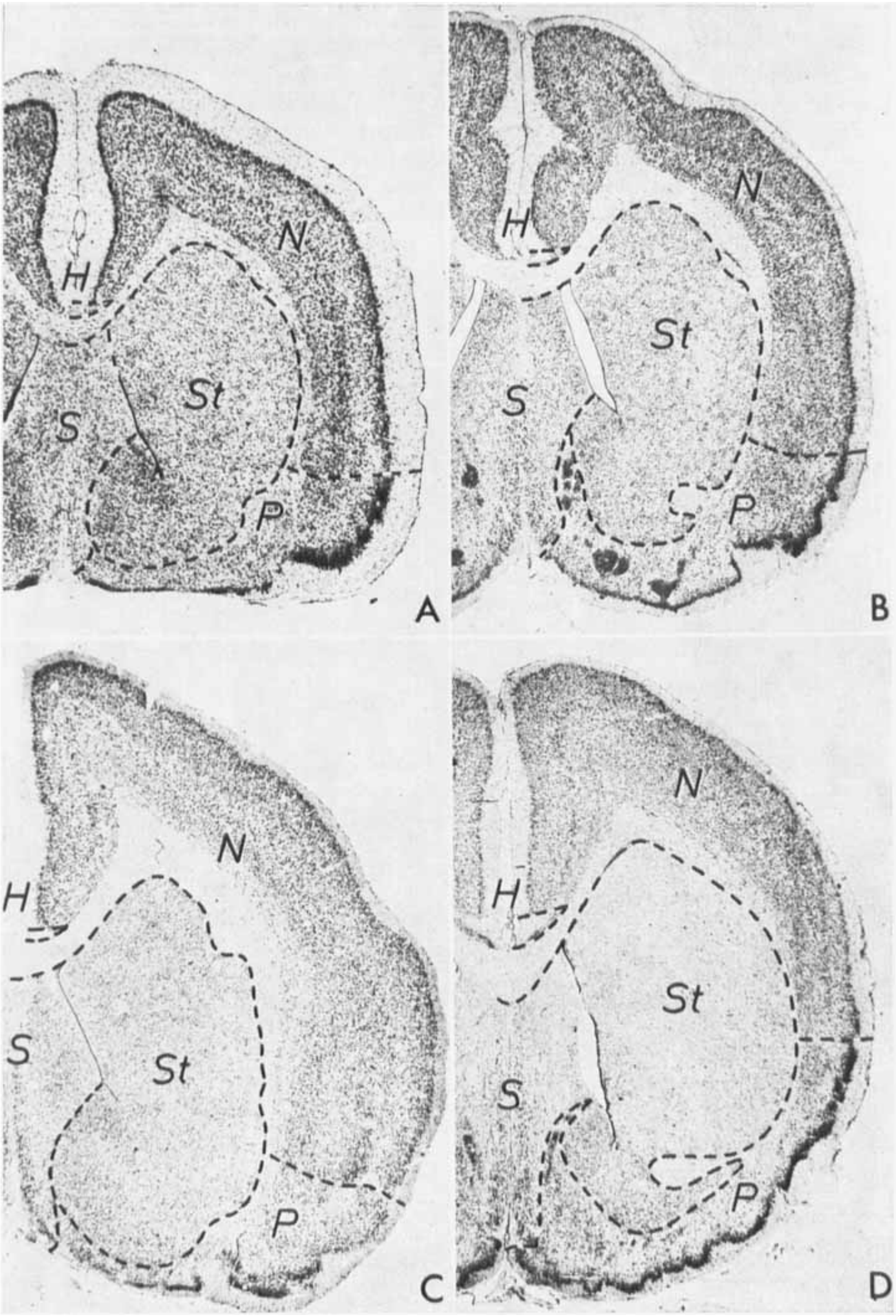

Fig. 1. Frontal sections through the region of the septum. Section thickness 10 μ, Cresylviolet. A *Asellia tridens* P 106, section 335, x 21.4. B *Artibeus jamaicensis* P 98, section 412, x 12.4. C *Noctilio leporinus* P 111, section 305, x 12.4. D *Cynopterus horsfieldii* P 103, section 440, x 12.4. Abbreviations: H Hippocampus, N Neocortex, P Palaeocortex, S Septum, St Striatum

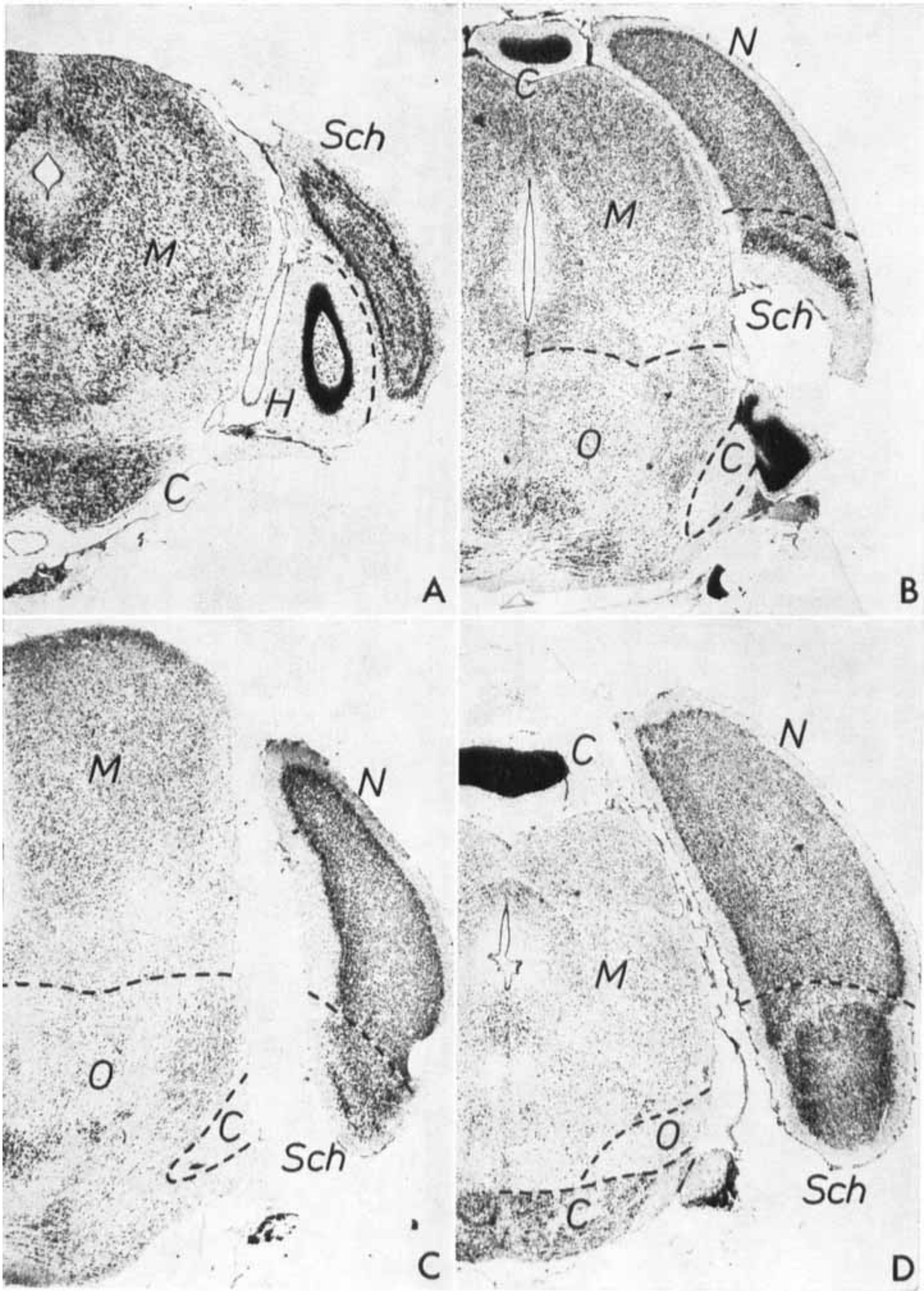

Fig. 2. Frontal sections through the caudal end of the cerebral hemispheres. Section thickness 10 μ, Cresylviolett. A *Asellia tridens* P 106, section 632, x 20.0. B. *Artibeus jamaicensis* P 98, section 892, x 11.9. C *Noctilio leporinus* P 111, section 785, x 11.5. D *Cynopterus horsfieldii* P 103, section 945, x 11.6. Abbreviations: C Cerebellum (Metencephalon), H Hippocampus, M Mesencephalon, N Neocortex, O Medulla oblongata, Sch Schizocortex. – Since it is almost impossible to separate the substantia reticularis of the oblongata from that of the mesencephalon, we have included the latter into the medulla oblongata. We also include the whole nucleus of the lemniscus lateralis into the medulla oblongata.

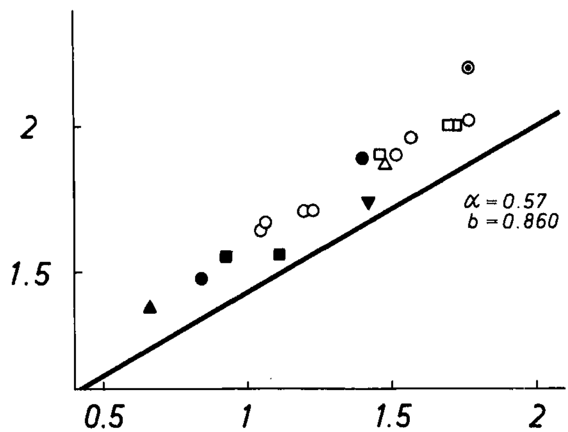

Fig. 3. Volumes of the medulla oblongata in relation to body weight on a double logarithmic scale. The base of reference is the line through the point of the mean values for the basal Insectivores with the most probable gradient for the medulla oblongata, which comes close to 0.57. This value was obtained through the comparison of as many related groups as possible in all material so far investigated (Insectivora, Primates, Chiroptera). It is the expression of the most typical size relation between volume of the medulla oblongata and body weight. α = slope of the line, b = point of intersection with the ordinate

structure of a certain species is larger than the corresponding structure in a typical basal Insectivore of the same body weight. The progression indices are indicative values, which, although not precise, are certainly sufficient for the type of comparison made here. – 5. Arrangement of the various progression indices in scales and comparisons (figs. 7–17). – Some further comments about the methods used are contained in the captions to the figures.

The following parts of the brain were studied: bulbus olfactorius, palaeocortex + amygdaloid complex, septum, striatum, schizocortex (= regio entorhinalis + praesubicularis), hippocampus, neocortex, diencephalon, mesencephalon, cerebellum and medulla oblongata.

In two cases (*Asellia tridens* and *Glossophaga soricina*) we have investigated two brains of the same species to get some indication of the degree of intraspecific variation. This is, in general, low and the individual progression indices of the various brain components are similar. The highest variations were found in *Asellia* for the septum (89–102, mean = 95) and in

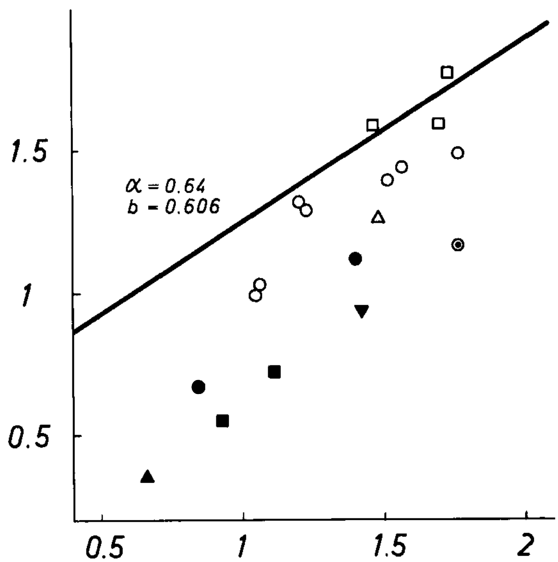

Fig. 4. Volumes of the olfactory bulb in relation to body weight on a double logarithmic scale. For explanation see fig. 3.

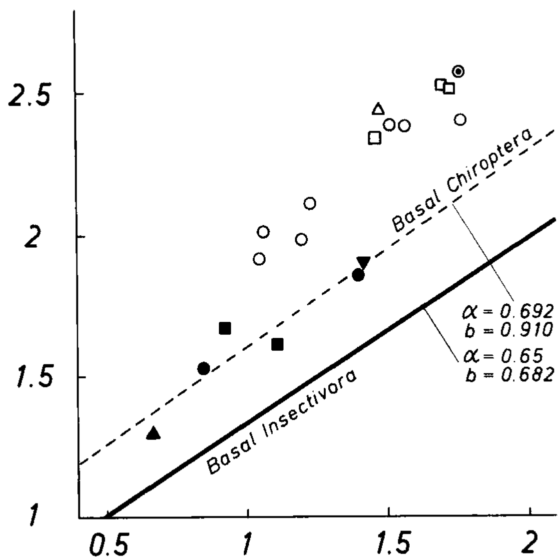

Fig. 5. Volumes of the neocortex in relation to body weight on a double logarithmic scale. For explanation see fig. 3. A regression line is also drawn for the "Basal Chiroptera".

Table 1

Absolute volume of the various structures or complexes (in mm³)

	Medulla oblongata 1	Cerebellum 2	Mesencephalon 3	Diencephalon 4	Bulbus olfactorius 5	Palaeocortex + NA 6	Septum 7	Striatum 8	Schizocortex 9	Hippocampus 10	Neocortex 11	Telencephalon (total) 12	Total brain 13	Body weight g 14
Rhinolophus hipposideros	24.3	32.5	16.9	13.9	2.3	11.1	2.1	6.9	5.9	14.8	19.7	62.7	150.3	4.6
Myotis myotis	77.6	86.2	44.1	46.6	13.2	42.9	7.2	22.0	20.9	36.5	72.0	214.6	469.1	25
Myotis nattereri	30.1	37.5	20.0	17.4	4.7	21.1	3.4	10.0	8.5	21.6	33.7	102.9	207.9	7
Chaerophon leucostigma	55.6	69.6	39.5	28.1	8.7	35.5	5.6	15.0	9.6	19.7	80.3	174.5	367.3	26.5
Asellia tridens	36.4	35.0	25.5	20.4	5.2	18.5	3.6	12.7	8.0	18.7	40.7	107.4	224.7	12.8
Hipposideros bicolor	35.5	42.4	25.6	18.8	3.6	18.1	3.1	12.4	5.7	14.7	46.4	103.9	226.1	8.4
Glossophaga soricina	44.1	73.4	26.4	30.5	9.9	26.4	5.4	17.9	12.6	30.1	82.3	184.5	358.8	11.2
Phyllostomus discolor	79.8	137.6	58.9	75.7	24.7	75.8	10.5	44.4	25.3	60.9	242.2	483.9	835.9	33
Carollia perspicillata	51.8	71.3	37.5	39.9	20.8	41.7	6.8	21.3	15.8	38.9	96.2	241.4	441.9	16
Sturnira lilium	51.3	74.1	30.5	42.8	19.7	41.5	7.1	31.2	16.9	43.5	128.6	288.6	487.3	17.1
Vampyrops helleri	46.9	83.0	32.0	33.1	10.8	28.6	4.5	16.0	9.0	27.2	102.9	199.0	393.9	11.6
Artibeus jamaicensis	91.5	156.2	62.7	80.3	27.8	67.0	10.3	47.2	26.6	59.5	241.7	480.0	870.7	37.5
Artibeus lituratus	104.1	202.9	76.1	88.7	31.0	71.9	11.2	48.9	35.6	62.5	249.8	510.9	982.6	58.7
Desmodus rotundus	74.0	156.7	45.5	68.2	18.0	50.7	6.8	43.5	22.6	29.9	273.0	444.6	788.9	30
Noctilio leporinus	157.9	200.6	90.1	108.7	14.8	83.6	12.8	73.1	26.0	48.7	372.3	631.4	1188.7	58.5
Cynopterus brachyotis	80.2	115.1	46.3	84.8	37.9	63.1	14.7	41.2	24.8	79.6	216.2	477.5	803.9	29
Cynopterus horsfieldii	100.8	150.7	76.1	118.2	58.5	87.9	20.8	77.1	39.3	92.7	322.0	698.1	1143.9	53
Eonycteris spelaea	99.2	151.0	65.5	116.1	38.7	79.2	18.3	66.1	38.4	96.1	330.5	667.2	1099.1	50

The data are net values, that is they must be referred to *pure tissues*: ventricles, meningeal membranes and nerves are excluded. These values are corrected to the fresh standard brain volume, which was calculated from the standard weights given by PIRLOT and STEPHAN (1970) by dividing these weights by the specific brain weight (= 1.036). Thus the factor of correction (C) is:

$$C = \frac{\text{volume of standard fresh brain}}{\text{serial section volume of individual brain}}$$

The schizocortex (Rose) comprises the regions located between the caudal and ventral hippocampus and the neocortex, that is the regio entorhinalis, regio perirhinalis and regio praesubicularis.

Glossophaga for the cerebellum (338–387, mean = 363), striatum (189–219, mean = 204) and the hippocampus (156–184, mean = 170). The greatest deviations of single values in % from the means for these structures are 7.4, 6.9, 7.4 and 8.2 %, respectively. Thus we expect, that in future investigations of broader materials the progression indices here given will not be changed by more than 10 %. Similar results were found in more comprehensive materials of primate brains (STEPHAN et al. 1970). The type of comparison here presented does not require a finer degree of accuracy. With two exceptions we have therefore investigated only one brain of each species – though as many species as possible.

Results

All volumes, measured in mm^3 are given in table 1.

Percentage composition of the brain

The relative percentage composition of the total brain and of the telencephalon is shown in tables 2 and 3. This is usually done as a matter of routine and first indication. Often thought, such data remain the only information about size in the reference literature. However, although the differences between the separate species are important, we will not discuss these figures further, or carry out a comparison of the various groups based upon them, because they do not provide precise information about the direction and the magnitude of the size modifications. Indeed, they do not take into consideration the variations in the total brain-size and in body weight. Our comparisons and discussions will therefore be based on the allometry formula.

Table 2

Percentage composition of the total brain

	Medulla oblongata 1	Cerebellum 2	Mesencephalon 3	Diencephalon 4	Telencephalon 5
Rhinolophus hipposideros	16.2	21.6	11.2	9.2	41.8
Myotis myotis	16.5	18.4	9.4	9.9	45.8
Myotis nattereri	14.5	18.0	9.6	8.4	49.5
Chaerophon leucostigma	15.1	19.0	10.7	7.7	47.5
Asellia tridens	16.2	15.6	11.4	9.1	47.8
Hipposideros bicolor	15.7	18.7	11.3	8.3	45.9
Glossophaga soricina	12.3	20.4	7.4	8.5	51.4
Phyllostomus discolor	9.6	16.5	7.0	9.0	57.9
Carollia perspicillata	11.7	16.1	8.5	9.0	54.6
Sturnira lilium	10.5	15.2	6.3	8.8	59.2
Vampyrops helleri	11.9	21.1	8.1	8.4	50.5
Artibeus jamaicensis	10.5	17.9	7.2	9.2	55.1
Artibeus lituratus	10.6	20.7	7.7	9.0	52.0
Desmodus rotundus	9.4	19.9	5.8	8.6	56.3
Noctilio leporinus	13.3	16.9	7.6	9.1	53.1
Cynopterus brachyotis	10.0	14.3	5.8	10.6	59.4
Cynopterus horsfieldii	8.8	13.2	6.7	10.3	61.0
Eonycteris spelaea	9.0	13.7	6.0	10.6	60.7

Table 3

Percentage composition of the telencephalon

	Medulla oblongata 1	Cerebellum 2	Mesencephalon 3	Diencephalon 4	Telencephalon 5
Rhinolophus hipposideros	16.2	21.6	11.2	9.2	41.8
Myotis myotis	16.5	18.4	9.4	9.9	45.8
Myotis nattereri	14.5	18.0	9.6	8.4	49.5
Chaerophon leucostigma	15.1	19.0	10.7	7.7	47.5
Asellia tridens	16.2	15.6	11.4	9.1	47.8
Hipposideros bicolor	15.7	18.7	11.3	8.3	45.9
Glossophaga soricina	12.3	20.4	7.4	8.5	51.4
Phyllostomus discolor	9.6	16.5	7.0	9.0	57.9
Carollia perspicillata	11.7	16.1	8.5	9.0	54.6
Sturnira lilium	10.5	15.2	6.3	8.8	59.2
Vampyrops helleri	11.9	21.1	8.1	8.4	50.5
Artibeus jamaicensis	10.5	17.9	7.2	9.2	55.1
Artibeus lituratus	10.6	20.7	7.7	9.0	52.0
Desmodus rotundus	9.4	19.9	5.8	8.6	56.3
Noctilio leporinus	13.3	16.9	7.6	9.1	53.1
Cynopterus brachyotis	10.0	14.3	5.8	10.6	59.4
Cynopterus horsfieldii	8.8	13.2	6.7	10.3	61.0
Eonycteris spelaea	9.0	13.7	6.0	10.6	60.7

Comparisons based on the allometry formula

The progression indices obtained by applying the allometry formula (cf. material and method) were arranged on a scale for each separate structure. As is the case in the Primates (STEPHAN 1967c), the neocortex is also in Chiroptera the most progressive structure (fig. 6 and table 4). This is clearly indicated by the fact, that both overall maximum (623) and the average value of the progression (376) are the highest among

Table 4

Data on progression indices in bats

	mean 1	max. 2	min. 3
Neocortex	376	623	152
Cerebellum	291	400	158
Mesencephalon	228	282	190
Diencephalon	225	327	114
Striatum	217	341	101
Schizocortex	188	255	93
Hippocampus	155	262	68
Medulla oblongata	150	214	118
Septum	144	238	95
Palaeocortex + NA	80	112	48
Bulbus olfactorius	56	114	21

all structures. We therefore have used it as a criterion for evaluating the evolutionary level reached by each particular species (for more detailed justification see p. 226–227). The importance of this highest site of integration in evaluating the degree of cerebral differentiation was also pointed out by WIRZ (1950) and SCHNEIDER (1957).

a. Neocorticalization (fig. 7)

With a mean at 376, the range of variation goes from 152 (*Rhinolophus*) up to 623 (*Desmodus*). At the lower end of the scale, we find representatives of the Rhinolophids (152), Vespertilionids (185 and 198), Molossids (198) and Hipposiderids (161 and 242). In these families, the neocortex is between 1.5 and 2.5 times larger than in basal Insectivores of equal body-size. They make a comparatively well defined inferior group. An intermediate group is formed by the Phyllostomatids among which the indices vary from 330 to 519 with a mean value of 408. The lowest index among the 7 species studied from this family belongs to *Carollia perspicillata* and the highest to *Phyllostomus discolor*. The other species lie at regular intervals in between. The Pteropids overlap slightly with the upper portion of the Phyllostomatid range. Three species of this group were available and they have indices ranging from 504 to 541, thus showing little variation. Their mean lies at 517 that is to say that, on the average, the Pteropids possess a neocortex 5.2 times larger than that of basal Insectivores of the same bodysize. In *Noctilio leporinus* the corresponding value is 5.5, which is very close to the upper limit of the Pteropids. The vampire *Desmodus rotundus* is even more highly neocorticalized with an index of 623.

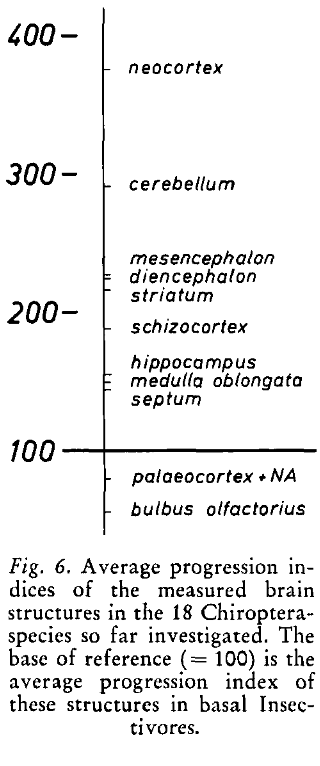

Fig. 6. Average progression indices of the measured brain structures in the 18 Chiroptera-species so far investigated. The base of reference (= 100) is the average progression index of these structures in basal Insectivores.

The rank-sequence based on the progression indices for the neocortex (fig. 7) can be interpreted as an "*Ascending Chiroptera Scale*" (for further comments see p. 227). The scale shows (1) that similar neocorticalization occurs in different families and therefore is not useful in evaluating systematic relationship (as already found for encephalization), (2) that distinctly different levels of evolution have been maintained until recent times and (3) among which recent groups the most primitive representatives of Chiroptera are to be expected, as estimated from their degree of neocorticalization.

b. "Basal Chiroptera"

If we are to speak of a "basal"[2] group among Chiroptera (the high specialization of the whole order being evident) we must look for such a group among the members of those families, that are well separated from all others with respect to neocorticali-

[2] *Basal* is to understood in the sense of "occupying within the *recent* Chiroptera, concerning the characteristics of the brain, the lowest place". In the course of phylogenetic development the fossil relatives of these recent basal Chiroptera, which may have played a role in the evolution of progressive Chiroptera, evidently have been stages of passage. For these fossils the terms *layer* or *level* would be more appropriate.

zation at the lower end of the scale (fig. 7): Rhinolophids, Vespertilionids, Molossids and Hipposiderids. In our former encephalization study, carried out on a broader material, these families were already located at the bottom of the scale[3], where they were accompanied by some Emballonurids and *Noctilio labialis* from the Noctilionids. In contrast, *N. leporinus* was clearly more highly encephalized, so that, within one

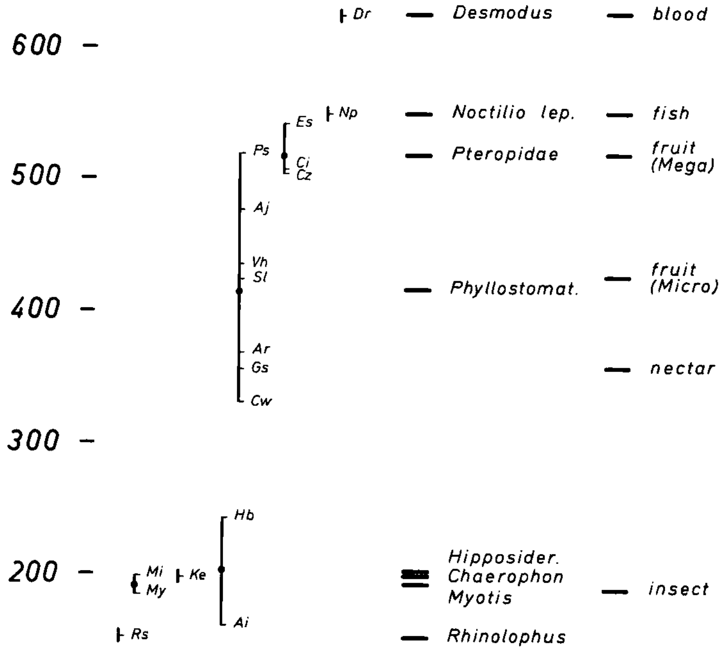

Fig. 7. Progression indices of the neocortex. They express, how many times the neocortex is larger than that of a typical basal Insectivore of equal body weight. On the left side (vertical columns) the different species and their range of variation within each family with the mean value of the indices shown by a dot. The family mean points form an ascending pattern from left to right. In the middle, the horizontal bars correspond to these means and the name of each family is given. On the right hand side the horizontal bars correspond to the means of the various dietary groups. The affiliation of the various species to these groups is given on page 219. – List of abbreviations: Ai *Asellia tridens*, Aj *Artibeus jamaicensis*, Ar *Artibeus lituratus*, Ci *Cynopterus horsfieldii*, Cw *Carollia perspicillata*, Cz *Cynopterus brachyotis*, Dr *Desmodus rotundus*, Es *Eonycteris spelaea*, Gs *Glossophaga soricina*, Hb *Hipposideros bicolor*, Ke *Chaerophon leucostigma*, Mi *Myotis nattereri*, My *Myotis myotis*, Np *Noctilio leporinus*, Ps *Phyllostomus discolor*, Rs *Rhinolophus hipposideros*, Sl *Sturnira lilium*, Vh *Vampyrops helleri*

family, "basal" as well as "progressive" species are represented. Corresponding relationships were also present in the order Insectivora (STEPHAN 1967b). Even within the Phyllostomatids, a family that includes outstandingly progressive types, some basal species can probably be found (*Mimon crenulatum*?). In the present investigation, neither this species nor *N. labialis* or the Emballonurids could be included.

Within the six species so far investigated, making up our basal group, the size increase of the neocortex with respect to enlarging body size is similar to what it is in the basal Insectivores (slope a = 0.69 versus 0.65, fig. 5). The median progression value

[3] A broader comparison based on both scales (neocorticalization and encephalization) will be carried out in a later paragraph (d), when the reasons for the existing differences in other brainparts can be discussed. These differences will be dealt with in the following paragraph (c).

for those "basal Chiroptera" (corresponding to the average distance between both straight lines in fig. 5) is 189, that is to say, the neocortex of the basal Chiroptera is on the average 1.9 times larger than that of the terrestrial basal Insectivores of equal body size. If one decides to base the comparison on the basal *Chiroptera* rather than on the basal *Insectivora,* one must then divide the indices given in the preceding paragraph

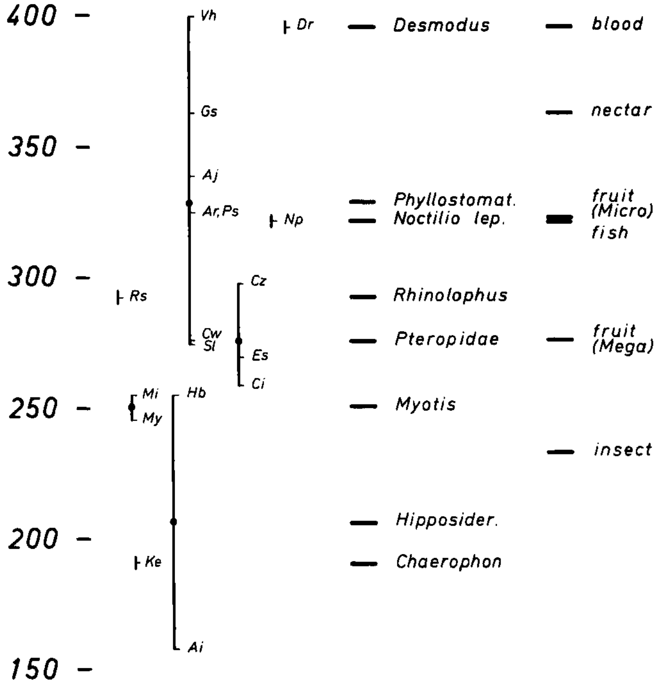

Fig. 8. Progression indices of the cerebellum. Explanations and abbreviations as in fig. 7

(a) by 1.9. In that case, the neocortex of the Phyllostomatids has an average value of 215, Pteropids 272, *Noctilio leporinus* 289 and *Desmodus* 329. The neocortex of *Demodus* is thus about 3.3 times larger than the average neocortex of the basal Chiroptera group.

We will limit ourselves here to the comparison with the basal *Insectivores.* But, with the aid of the mean progression values for the basal Chiroptera [4], one can in any case quickly calculate what indices can be obtained when the comparison is based on Chiroptera. The indices given hereafter and those of fig. 8 to 17 should therefore be divided separately for each structure by the mean index of that structure in the basal Chiroptera (table 4, column 1).

c. Comparisons of the progression indices of the various structures in the different systematic groups

We can now examine the behaviour of the various brain structures so far investigated in the ascending series of Chiroptera. The part of the encephalon which shows the next highest average progression (after the neocortex) is the cerebellum (mean =

[4] These correspond to the insect-eating Microchiroptera, as will be shown in a later paragraph (e) (page 219) and discussion (page 221).

291). Its generally large size in bats has already been referred to by WIRZ (1950). The cerebellum is followed by the mesencephalon (228), the diencephalon (225) and the striatum (217). After that come the schizocortex (188), the hippocampus (155) and the septum (144), that is, the three structures belonging to the limbic system, and the medulla oblongata (150). The palaeocortex (80) and in particular the bulbus olfactorius

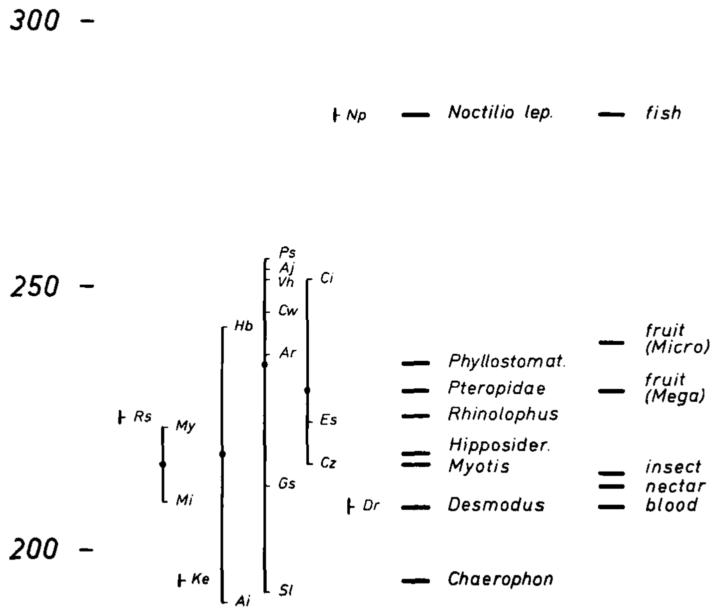

Fig. 9. Progression indices of the mesencephalon. Explanations and abbreviations as in fig. 7

(56) show, on the average, a regression or decrease in comparison with the basal Insectivores (fig. 6).

These average values give only a general indication of the trend and intensity of size modification with respect to basal Insectivores. The separate values for each systematic group or species vary greatly (see table 4) and they will be analyzed more accurately hereafter, as was done above for the neocortex.

We would like to re-emphasize that our indices provide standard and indicative values of the progression, but are not statistically precise.

Cerebellum (fig. 8). The progression of the cerebellum varies, at first sight, in a wide range, from 158 in *Asellia* to 400 in *Vampyrops.* Next to *Asellia,* only *Chaerophon* has an index inferior to 200, while the indices for all the other species so far studied vary between 245 und 400, being thus contained within a comparatively small range. This range limitation may point to a great importance of the cerebellum in all bats (see further).

But then the question arises, as to why *Chaerophon* and *Asellia* stand so far below the others. Furthermore, *Asellia* itself is very widely separated from the other species (*Hipposideros bicolor*) of the same family (Hipposideridae), which we have investigated. No error in the measurements can explain this since, as already said, we have studied two brains of *Asellia* and found the size of the various parts in full agreement with one another. The progression index for the cerebellum was 153 in the first and 162 in the second instance.

There is no parallelism in comparing the progression sequence for the cerebellum with the "ascending Chiroptera-scale". Cerebellum and neocortex are therefore not strongly correlated with one another as to their size. In particular, the rank sequences among the "basal Chiroptera" differ greatly on the two scales. Furthermore, in the Pteropids, the cerebellum is comparatively little developed. While the bats of this family stand near the top of the scale as regards the neocortical development, they only have a medium-sized cerebellum, together with such remote types as the *Myotis*

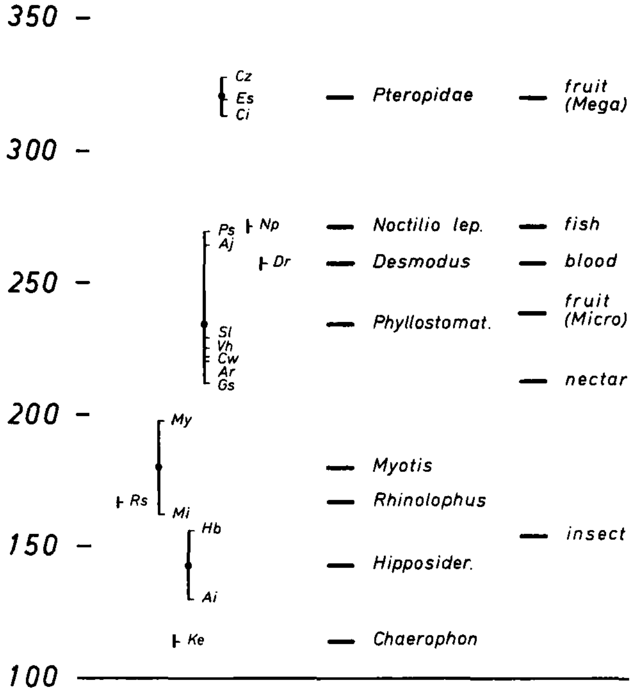

Fig. 10. Progression indices of the diencephalon. Explanations and abbreviations as in fig. 7

species, with *Hipposideros*, *Rhinolophus* and with the Phyllostomatids *Sturnira* and *Carollia*. The Phyllostomatids as a whole show a very great variability. With the two *Artibeus* species, with *Phyllostomus* and *Glossophaga*, they form (together with *Noctilio leporinus*) a median group and reach in *Vampyrops* their highest value in a region of the scale attained only by *Desmodus*, among the other species.

Mesencephalon (fig. 9). The third highest average progression is found in the mesencephalon (228), with a variation from 190 to 282. If one excludes *Noctilio leporinus*, a species which surpasses all others, the range is restricted to 190–255. Such a variation is unusually narrow and corresponds in magnitude to that found in the medulla oblongata (see below). This means, that all Chiroptera have achieved approximately the same progression, with regard to the size of the mesencephalon. It does not, however, imply that the composition of this complex is exactly the same in all species. A quantitative research into the anterior and posterior colliculi is now in progress (Baron, in preparation) and might well lead to interesting conclusions concerning the particular position of *Noctilio leporinus*.

The ascending sequence is here very much disturbed, i. e. the neocortex and the mesencephalon show no narrow size correlation.

Diencephalon (fig. 10). The diencephalon reaches an average progression value (225) similar to that of the mesencephalon, although its variation range is much greater. It goes from 114 in *Chaerophon* up to 327 in *Cynopterus brachyotis*. A lower group is represented by *Chaerophon*, the Hipposiderids, *Rhinolophus* and *Myotis*, a middle group by the Phyllostomatids, *Desmodus* and *Noctilio*, and the clearly highest group by the Pteropids. With the material on hand, we have found no overlap between those three groups. In the ascending sequence (i. e. always with reference to the neocorticalization scale, fig. 7), the diencephalon shows a clear trend to enlargement. There is, how-

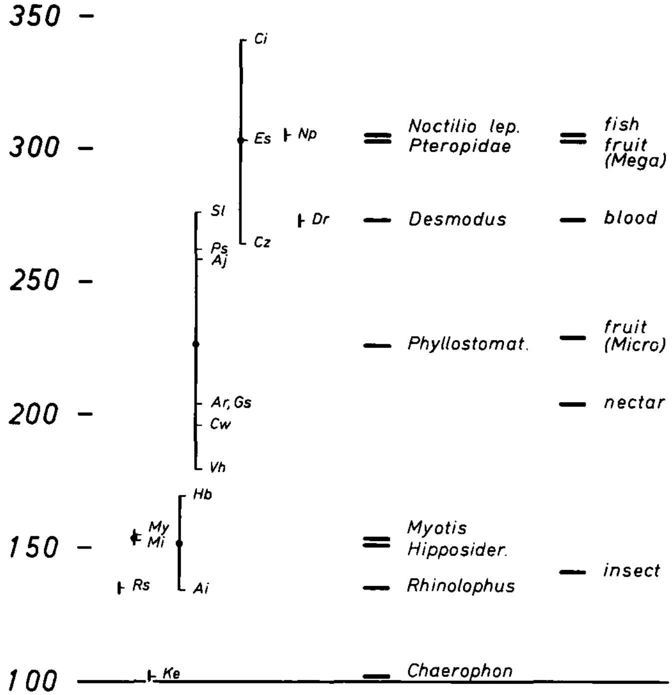

Fig. 11. Progression indices of the striatum. Explanations and abbreviations as in fig. 7

ever, no complete agreement between the two sequences since the diencephalon is comparatively small in *Noctilio* and *Desmodus* and/or comparatively large in the Pteropids. Corresponding relations appear if one compares *Chaerophon* and the Hipposiderids (small) with *Rhinolophus* and *Myotis* (large).

Striatum (fig. 11). The average progression of the striatum (217) is almost as high as that found in the diencephalon and the mesencephalon. The variation, however, is still greater than in the case of the diencephalon, ranging from 101 in *Chaerophon* to 341 in *Cynopterus horsfieldii*. The maximum attained (341) is the third highest, coming after the neocortex (623) and the cerebellum (400) (table 4, column 2). The lower group is made of *Chaerophon*, *Rhinolophus*, the Hipposiderids and *Myotis*; the middle group of the Phyllostomatids and the upper group of the Pteropids and *Noctilio*. Here, *Desmodus* lies in the zone of overlap between middle and upper groups. The "ascending Chiroptera scale" is nearly unchanged. Only *Chaerophon* and *Desmodus* show, in comparison with their neocorticalization, a small striatum. There are obviously narrow size-relationships between striatum and neocortex which were also observed in Primates (STEPHAN 1967c).

Schizocortex (fig. 12). The schizocortex has an average progression index of 188 with a variation range from 93 to 255. Both *Myotis* species show comparatively high values. In contrast, *Vampyrops* (a Phyllostomatid) is very low and so appears *Desmodus*, if one refers to the very high neocorticalization of this species. The Hipposiderids show very little progress while *Chaerophon* is even characterized by a slight regression. In these forms, the schizocortex is practically no larger than in the basal Insectivores of equal body size. The Pteropids are evidently the bats with the largest schizocortex and, in this respect, they occupy a top position without any overlap. One cannot detect, in the schizocortex, a clear ascending trend that would follow the Chiroptera scale. Schizocortex and neocortex vary therefore largely independently from one another.

Hippocampus (fig. 13). The hippocampus has a mean progression of 155 with a variation from 68 to 262. In *Chaerophon*, it is clearly lower than the mean value for the basal Insectivores (i. e. 100), which indicates a reduction or regressive development of that structure in that bat. Although a final confirmation of this observation could

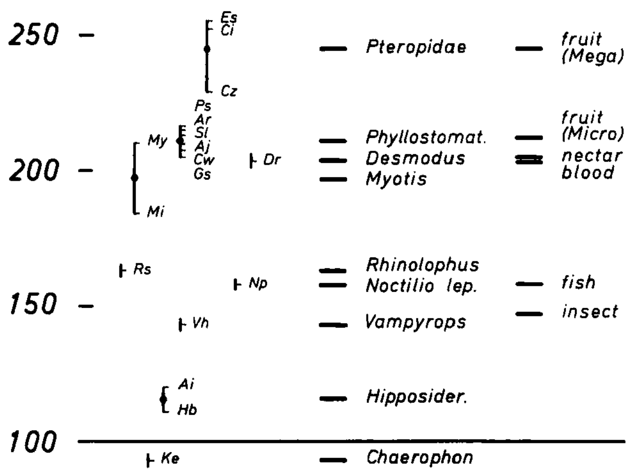

Fig. 12. Progression indices of the schizocortex (Regio entorhinalis, perirhinalis and praesubicularis). Explanations and abbreviations as in fig. 7

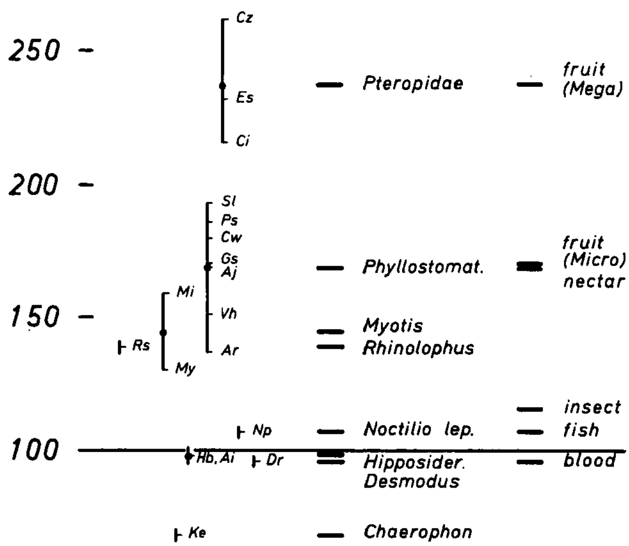

Fig. 13. Progression indices of the hippocampus. Explanations and abbreviations as in fig. 7

only be provided by investigation of more specimens of the same species, it seems that the regressive development of the hippocampus is likely to have occurred in at least a number of bats, since three other species (*Desmodus*, *Asellia*, *Hipposideros*) show indices below 100 and *Noctilio* lies very little above that value. The largest hippocampus is found in the Pteropids.

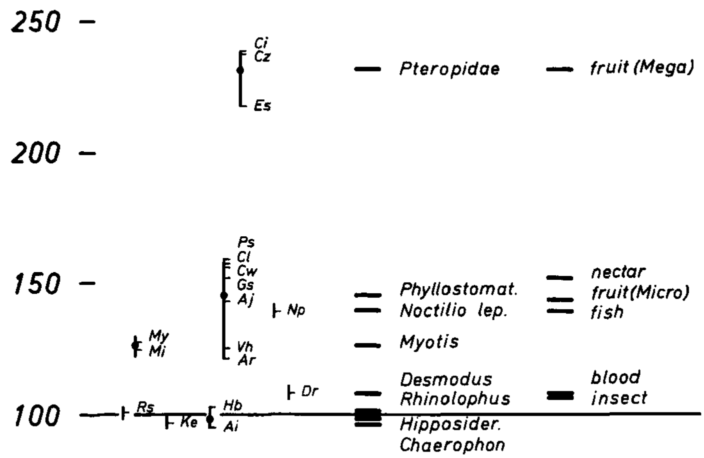

Fig. 14. Progression indices of the septum. Explanations and abbreviations as in fig. 7

Septum (fig. 14). With respect to the septum, the Pteropids stand even further above the other groups. The septum has an average progression index of 144 with a range from 95 to 238. It is comparatively small in *Chaerophon*, the Hipposiderids, *Rhinolophus* and *Desmodus*, with indices close to 100. Here again, a slight regressive development probably occurs in a number of species.

The progression scales for the last three structures here discussed (schizocortex, hippocampus and septum) show great similarities, including in the following characteristics: greatest development in the Pteropids; smallest size in *Chaerophon* and in the Hipposiderids; intermediate position of the Phyllostomatids and *Myotis*. These three structures belong to the so-called "limbic system" and they are functionally interrelated. No distinct progressive trend affecting any of these structures is visible on the ascending scale. This is also obvious from the fact, that the septum and the hippocampus show particularly low indices in *Desmodus* and *Noctilio*, the two species which, for their neocorticalization, are at the top of the scale.

Medulla oblongata (fig. 15). The progression scale for the medulla oblongata is in many respects similar to that for the mesencephalon. In both cases, the index for *Noctilio* is remarkably high, those for *Asellia* and *Chaerophon* particularly low. If one deletes those rather aberrant species, both strutures show but an exceptionally low variability. The average progression for the medulla (150) is however much lower than for the mesencephalon (228). One cannot determine any definite trend of size-development in the ascending Chiroptera scale.

All the structures so far discussed were, on the average, progressive in comparison with the basal Insectivores. If we compare their mean progression with that found in Primates (data from Stephan and Andy 1969), several remarkable differences can be brought out. These pertain not only to the magnitude of the progression, which is in most cases clearly higher in Primates, but also to the rank-sequences of the structures themselves. These differences in rank are surely significant in relation to differences in the locomotion. The neocortex shows the strongest progression in both Primates and Chiroptera. But, while in the Primates the second place is taken by the striatum, the third by the diencephalon and the fourth only by the cerebellum, in the Chiroptera the cerebellum clearly supplants the striatum and the diencephalon in its progression. The

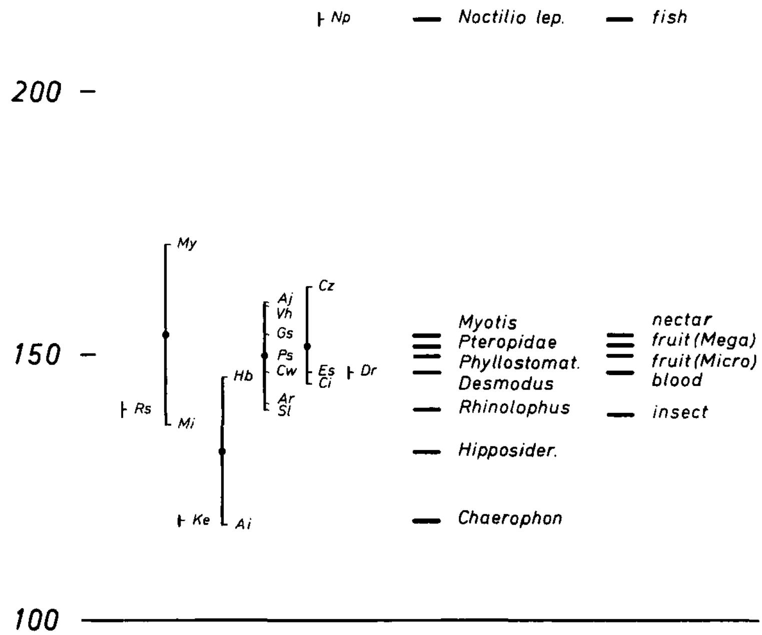

Fig. 15. Progression indices of the medulla oblongata. Explanations and abbreviations as in fig. 7

striatum itself is superseded by the mesencephalon, a structure which is evidently less progressive in the Primates.

In this connection, it is worth noting two facts: (1) the striatum and the diencephalon are after the neocortex the structures with the strongest progression in Primates; (2) they are also those demonstrating a markedly progressive trend on the ascending Chiroptera scale, such a trend not being observed in any other structure.

The following two structures are (on the average) regressive.

Bulbus olfactorius (fig. 16). The bulbus olfactorius with a mean index of 56, on the average attains only $^3/_5$ of its size in the basal Insectivores. Its variation ranges from 21 to 114. The strongest regression of the bulbus is found in *Rhinolophus*, in the Hipposiderids, in *Noctilio leporinus* and in *Chaerophon* (indices from 21 to 27); it is less marked in *Myotis* (33 and 42), *Desmodus* (51) and the Phyllostomatids (52 to 87), while in the Pteropids there is practically no regression (indices from 78 to 114). In the latter, the bulbus olfactorius is on the average as large as in the basal Insectivores (i. e. = 100).

Figure 16 gives the impression that there exists a size-relationship between olfactory bulb and neocortex, so that the former becomes larger as the latter increases. But this tendency is not observed in *Desmodus* and *Noctilio leporinus*, and in the Primates the trend is reversed: the olfactory bulb becomes smaller as the neocortex enlarges. The partial parallelism in fig. 16 is accidental.

Palaeocortex + amygdaloid complex (fig. 17). The quantitative relationships observed in this structural complex are similar to those in the olfactory bulb but the regression is, as a whole, less marked. This is mainly due to the fact that the amygdaloid nucleus is included in this complex. Contrary to the true olfactory regions of the palaeocortex, this structure is probably progressive, at least in some of its components.

d. Comparison between neocorticalization and encephalization

Our study of the encephalization i. e. of the total brain size (PIRLOT and STEPHAN 1970) was intended to provide a preliminary reference basis for estimating the level of evolution attained by the various species of bats. But, since the brain as a whole does not enlarge proportionately (i. e. to the same extent in each of its parts), the level of evolution will certainly be best appreciated in that structure which shows the

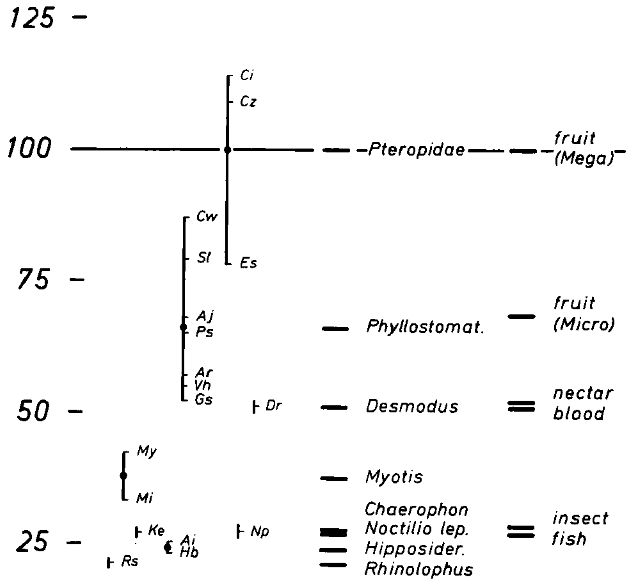

Fig. 16. Progression indices of the olfactory bulb. Explanations and abbreviations as in fig. 7

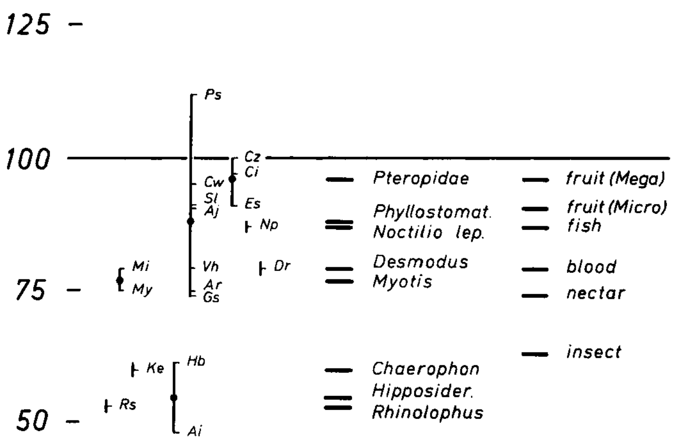

Fig. 17. Progression indices of the palaeocortex + amygdaloid complex. Explanations and abbreviations as in fig. 7

greatest size increase. That is the neocortex. We have therefore considered, in the Chiroptera, the scale of increasing neocorticalization as representing the best possible quantitative criterion upon which an "ascending Chiroptera scale" can be based. By reference to that scale, we can test if and to which extent the encephalization permits conclusions as to the level of evolution.

Indeed, if we compare encephalization with neocorticalization, we observe an obvious and broad agreement between the two. However, there are also differences

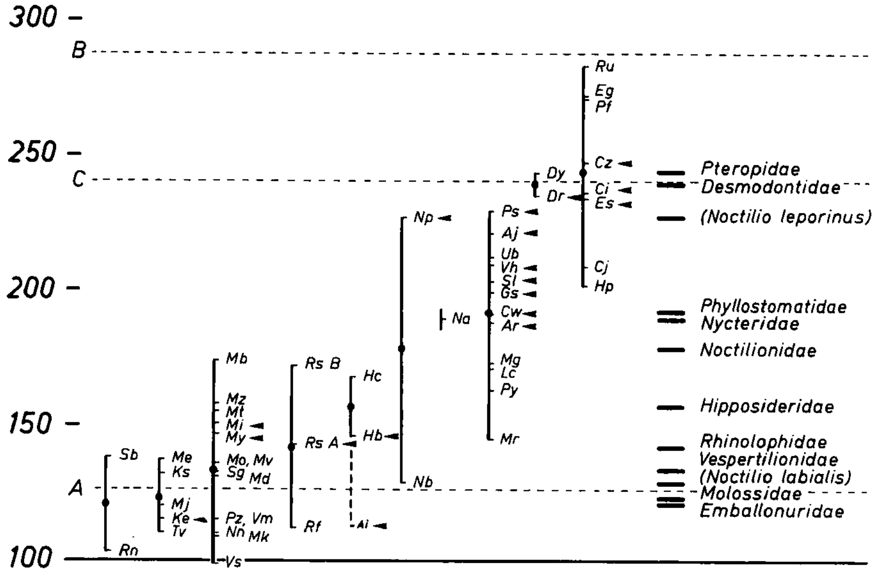

Fig. 18. Indices of encephalization arranged according to the families (from Pirlot and Stephan 1970). – Species investigated in the present study are indicated by pointers. *Asellia tridens* (Ai) was added without changing the mean of the Hipposideridae.

in the rank-position of single species. These differences are not only due to intraspecific variations, but rest upon some important size differences present in non-cortical brain components. One good example among others is here the particular position of *Desmodus*. This species is clearly more highly neocorticalized than the Megachiroptera so far investigated. On the encephalization scale, however, both groups are located on the same level. In cases of equal encephalization but larger neocorticalization, other structures must be smaller in *Desmodus* than in the Pteropids. These, according to the above comparisons (fig. 8–17), are mainly the mesencephalon, the structures of the olfactory (bulbus olfactorius and palaeocortex) and the limbic systems (septum, hippocampus and schizocortex).

Similar shifts in the two progressions scales (encephalization – neocorticalization) become also visible by comparing highly evolved Chiroptera with lower Primates. In their encephalization (Pirlot and Stephan 1970), the Chiroptera reach into the Prosimian zone (the lower border of the latter is indicated by the dotted line C in Fig. 18), as previously noted by Mangold-Wirz (1966). From the species here investigated, the three Pteropids and *Desmodus* reach their lowest limit (fig. 18). In contrast, in the neocorticalization (fig. 7), an obvious gap can be observed: the highest neocorticalized among the bats (*Desmodus* with an index of 623) lies clearly lower than the lowest of the Prosimians (724 in *Urogale everetti* or, if one excludes the Tupaiids, 812 in *Lepilemur ruficaudatus*). The larger size of the neocortex in the low Prosimians is compensated in Chiroptera by a greater development of the cerebellum (*Desmodus*) and also olfactory structures (Pteropids). Regarding the total brain the result is a similar encephalization.

The general agreement between both progression scales confirms the value of the encephalization as a useful rough guideline for the level of evolution attained by a species. However, as the above examples show, an accurate judgment can only be made after a more detailed analysis.

e. Comparisons of the progression indices in the various dietary types

In our investigation of the total brain (PIRLOT and STEPHAN 1970), we have already demonstrated a clear interdependence between brain size and dietary regime. PIRLOT (1969) has adopted the following classification: insect-eaters, nectar- (and pollen-) feeders, frugivorous Microchiroptera, fish-eaters, blood-suckers and frugivorous Megachiroptera. Within the Megachiroptera we have not yet distinguished between the more nectarivorous (*Eonycteris*) and the more frugivorous (*Cynopterus*) species, since the material is still restricted and the representatives of both groups are very similar both in encephalization and in progression of the various structures. We have had at our disposal no specimen representing a true carnivorous diet, but we hope to be able to consider this type of regime in a later study.

We include among the insect-eaters those species which are exclusively insectivorous and catch their prey mainly on the wing, but not those which eat insects alongside other foods. To this latter category belong undoubtedly the nectar-feeders and the fish-eaters, and perhaps also some of the frugivorous Microchiroptera.

Among the species here studied, the insect-eaters are represented by the families Rhinolophidae, Hipposideridae, Vespertilionidae and Molossidae (6 species: Rs, Hb, Ai, Mi, My, Ke) the nectar-feeders by *Glossophaga soricina* (family Phyllostomatidae; one species: Gs), the frugivorous Microchiroptera by the bulk of the Phyllostomatids (6 species: Vh, Aj, Ar, Ps, Cw, Sl), the fish-eaters by *Noctilio leporinus* (Np) the blood-suckers by *Desmodus rotundus* (Dr), and the frugivorous Megachiroptera by the Pteropids (3 species: Cz, Ci, Es).

In figures 7 to 17, the mean progression index for each *family* is indicated in the middle, for each *feeding habit type* on the right-hand side. The variation range of the indices within the latter can be estimated from the above key to the systematic classification of their representatives as indicated by the symbols on the left hand side of the graphs.

Evidently our "basal Chiroptera" are identical with the insectivorous bats. The material on hand is still too limited to permit a decision as to whether all insectivorous bats (according to our above definition) do belong to the basal Chiroptera or if some progressive species are also included in this dietary group. From our investigation of the encephalization, such progressive types, if they exist, may be represented by the Nycterids.

This family possesses the highest encephalization among all insectivorous bats so far studied (fig. 18). It is, however, quite likely that the Nycterids do not correspond perfectly to the typical insectivorous bats, since they seem to feed to some extent upon non-flying Arthropods. Their special behaviour is described by GRIFFIN (1958, p. 250).

Neocortex. The insect-eaters have by far the lowest neocorticalization. The mean value is 189 with a range from 152 to 242, in which *Rhinolophus* and *Asellia* present the lowest and *Hipposideros* the highest figures, while *Chaerophon* and the two *Myotis* occupy an intermediate position. There is no overlap with other dietary types but it must be remembered that our 18 species form a comparatively restricted basis. However, the overlap was very small also with respect to the encephalization (Pirlot and Stephan 1970). The frugivorous and nectar-feeding species, which all belong to the Phyllostomatids, form the next group in the ascending sequence. The nectar-feeder is situated near the lower end of the variation range of this group, as was already the case with the encephalization. Its neocorticalization index is 356, whereas the indices of the frugivorous species vary from 330 to 519 with their highest value in *Phyllostomus discolor*. For the encephalization also, this last species showed the highest figure among all frugivorous Microchiroptera which reach into the range of variation of the Megachiroptera. The latter have a mean index of 517 with a range of variation from 504 to 541. This narrow range will probably be widened by the study of more species. The fish-eater *Noctilio leporinus* is even more highly neocorticalized (549). The highest degree of neocorticalization, found in our material, belongs to the blood-sucker *Desmodus rotundus*, with an index of 623. The neocortex of this vampire is thus over six times the size of that of a basal Insectivore with the same body weight.

The position of the non-neocortical structures in the above sequence (i. e. insect-eaters, nectar-feeders, frugivorous Microchiroptera, frugivorous Megachiroptera, fish-eaters, blood-suckers) will be hereafter briefly characterized.

Cerebellum. The cerebellum is particularly well developed in the vampires and the nectar-feeders, whereas it is of intermediate development in the frugivorous and piscivorous Microchiroptera. The comparatively smallest cerebellum is found in the Megachiroptera and in the insectivorous Microchiroptera. We will discuss these differences in progression later from a functional viewpoint. Compared with the neocorticalization, the low position of the Megachiroptera and the comparatively high position of the nectar-feeders are striking.

Mesencephalon. The mesencephalon is particularly large in the fish-eater compared to all other types. Some differences are found in the average values of the other types, but they are very slight and overlaps are extensive. The mesencephalon is comparatively small in some insect-eaters and in the frugivorous *Sturnira lilium*.

Diencephalon. The ascending sequence for the diencephalon is very much like that for the neocortex, excepting the vampire, which occupies not the top position but only third rank. The frugivorous Megachiroptera possess by far the largest diencephalon and the insect-eaters the smallest. No overlap was found between these two extreme groups and the neighbouring ones.

Striatum. The striatum presents the same ascending scale as the neocortex, again with the exception of the vampire. In this bat, the striatum is smaller than in the fish-eater, also smaller than the mean for the Megachiroptera, a peculiarity, which we also observed in the case of the diencephalon. There is much overlap between the various dietary types, but the insect-eating species clearly have the smallest striatum.

Schizocortex. The schizocortex reaches its largest size in the frugivorous Megachiroptera. It is greatly variable among the insect-eaters, where it appears to be small on the average. The fish-eaters and *Vampyrops helleri* (frugivorous Microchiroptera) have also a small schizocortex. However, the figure for *Vampyrops* needs confirmation because this cortical area of the available specimen was not perfectly intact.

Hippocampus and Septum. These two structures can be treated together as they show very similar trends. Their unusually low value in the vampires is surprising and will be commented upon, when we deal with the functional problems. The highest values are attained by the frugivorous Megachiroptera. The frugivorous Microchiroptera and the nectar-feeders occupy an intermediate position while the insectivorous species form the lowest group next to the vampire. The fish-eater is consistently located in the lower part of the scale.

Medulla oblongata. The medulla oblongata shows practically the same development in all dietary groups, excepting the fish-eater in which it is distinctly larger. We can here draw a parallel to the mesencephalon.

Bulbus olfactorius. The size of the olfactory bulb shows a clear relationship to the diet. This primary olfactory center is least developed in the fish- and insect-eaters, somewhat stronger in the nectar-feeder and the vampire, and obviously largest in the frugivorous species; even more so in the Megachiroptera than in the frugivorous Microchiroptera. This point will be further discussed below. In the palaeocortex, we found similar quantitative relationships, but the differences are less marked.

Discussion and Conclusions

1. Encephalic characteristics of the dietary types and related functional aspects

Insectivorous Microchiroptera (fig. 19a and table 5, column 1). The brains in this group are, in general, little progressive and must be described as primitive. In a series of structures, we see that the average size does not markedly exceed that found in corresponding structures of basal Insectivores. In opposition to this, the cerebellum and, to a smaller extent, the mesencephalon appear as strikingly progressive, even more so than the neocortex. Here, for the first time in all mammalian brains so far investigated (with exception of the brains of the Macroscelididae, Insectivora), we find that, in a progressive sequence of encephalic structures, it is not the neocortex but another part of the brain that shows the highest progression. This is certainly related to the exceptionally high specialization of Chiroptera. The capture of a flying insect presupposes an outstanding flight-ability. One of the morphological bases for this ability lies obviously in the cerebellum as the center for movement coordination, muscular tonus and body-balance. Functional requirements and size go hand in hand.

Within the sensory systems the ultra-sonic one seems to be most important to the insect-hunters. This is reflected in the high degree of progression of the medulla oblongata and mesencephalon, where the prominent centers of this system are represented. A comparative study of specific brain structures, which are part of the ultra-sonic system, is now in progress (BARON, in preparation).

In contrast, the olfactory system, which is so extraordinarily important in terrestrial insect-eaters (order Insectivora), looses its significance in the flying insect-hunters; it is in fact reduced to about one third of its original size in its primary center, the olfactory bulb. The lesser reduction of the palaeocortical-amygdaloid complex can certainly be explained by the fact that some non-olfactory centers are included therein (baso-lateral group of the amygdala).

Summarizing: The progression of the neocortex is comparatively modest in the insectivorous bats. Thus, the development of higher centers of integration, which could give a certain plasticity to the functions of the nervous system, is evidently very little advanced. The overall construction of the brain in these bats appears to be highly specialized though extraordinarily primitive.

Nectarivorous Microchiroptera (fig. 19b and table 5, column 2). The only member of this group available to us was *Glossophaga soricina,* of the family Phyllostomatidae, and the sub-family Glossophaginae. Our results are therefore only provisional. Another restriction is, that *Glossophaga* does not apparently feed exclusively on nectar. According to WALKER (1964), it hunted for insects actively and, in captivity, insects were at times its favorite food. The question remains whether purely nectarivorous bats really exist.

In *Glossophaga* the progression of the cerebellum overrides considerably the value found for insectivorous species. This can again be related to a highly skilled flying ability: the species, beside hunting actively for insects, is able to hover in front of flowers like colibris do. Furthermore, in *Glossophaga*, the neocortex also is markedly developed, so that both structures are about equally progressive (fig. 19b).

With respect to many characteristics of brain development, *Glossophaga* seems to be a transitional type between purely insectivorous and truly frugivorous Microchiroptera (compare figs. 19a, b and c). Leaving aside the cerebellum, this is valid for all strongly progressive structures (neocortex, diencephalon and striatum) on the one hand, as also for the olfactory structures, on the other. Compared with insect-eaters, the olfactory structures show an increased size in relation to the use of vegetabel food.

Summarizing: Glossophaga appears as a very highly specialized form with clear progresses in its brain development, in contrast with the pure insect-eaters, which are little advanced by comparison.

Frugivorous Microchiroptera (fig. 19c and table 5, column 3). In the frugivorous Microchiroptera, the neocortex is even more strongly developed and has definitely superseded the progression of the cerebellum. The cerebellum itself is smaller than in

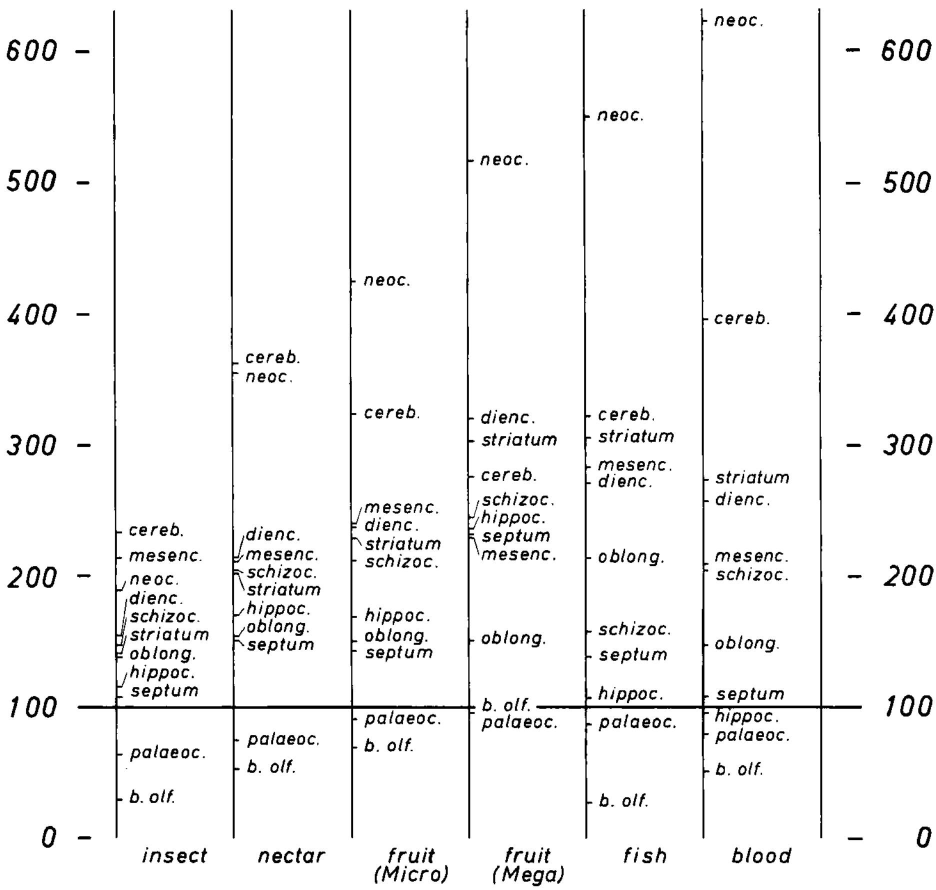

Fig. 19. Average progression indices of the measured brain structures in the various dietary groups. The base of reference is the average progression index of these structures in basal Insectivores, which we have inserted as being 100.

Glossophaga, a trend carried forth into the group of frugivorous Megachiroptera, and reflected in the lesser flight-ability of these animals. Nevertheless, the cerebellum is still

Table 5

Mean values of the progression indices in the various dietary groups

	insect 1	nectar 2	fruit micro 3	fruit mega 4	fish 5	blood 6
Neocortex (neoc.)	189	356	425	517	549	623
Cerebellum (cereb.)	233	363	323	276	322	396
Mesencephalon (mesenc.)	214	212	239	230	282	208
Diencephalon (dienc.)	154	212	238	320	271	257
Striatum (striatum)	141	204	229	303	305	273
Schizocortec (schizoc.)	147	205	212	245	158	204
Hippocampus (hippoc.)	115	170	169	237	107	96
Medulla oblongata (oblong.)	139	154	150	152	214	147
Septum (septum)	107	152	143	231	139	108
Palaeocortex + NA (palaeoc.)	63	74	91	96	87	79
Bulbus olfactorius (b. olf.)	29	52	69	100	27	51

In brackets the abbreviations as used in fig. 19.

better developed than in the insectivorous Microchiroptera. This may look strange at first, but can be related to the fact, that frugivorous species have maintained or recovered a certain ability to climb in trees.

With the stronger development of the neocortex, the diencephalon and the striatum have an increased emphasis in this group also. The mesencephalon is larger than in *Glossophaga* and the insect-eaters, which is possibly due to an increase of the anterior colliculi (visual relays). Here again, the work of BARON (in preparation) may yield further conclusions. The size of the eye is, in general, greater in the frugivorous than in the insectivorous species.

The centers of the olfactory system have increased considerably in comparison with those of the insect-eaters. The functional significance of this development is the adaptation to static vegetable food to be found and tested mainly by olfaction.

Summarizing: The frugivorous Microchiroptera form a group which (1) shows great progress in its encephalic development and (2) in opposition to the pure insectivorous types, suggests a weakening of the unilateral character of specialization: that is, a reduced dominance of a single sensory system and a widening of the locomotion spectrum (climbing habits).

Frugivorous Megachiroptera (fig. 19d and table 5, column 4). The enlargement of the olfactory centers is still more emphasized in frugivorous Megachiroptera (Pteropids). The progression indices for these centers reach here values which are typical of basal Insectivores, although the feeding-habits of both groups are entirely different. Quantitatively, the structures are similar in both cases; possible qualitative, histological differences are under investigation.

The clear differences in olfactory bulb size between the nectar-feeder *Eoncyteris* and the fruit-eaters of the genus *Cynopterus* is worth mentioning and parallels what was found for corresponding dietary groups in the Phyllostomatids. Generally, the nectar-feeders seem to have less developed olfactory structures than the fruit-eaters.

Neocortex, diencephalon and striatum have further increased but, in contrast, the cerebellum has not. The latter is clearly smaller than in nectarivorous and frugivorous Microchiroptera, taking only fourth place on the progression scale. This sequence resembles strongly that of the Primates (STEPHAN and ANDY 1969). The possession of a comparatively small cerebellum stands in good agreement with the lesser flying ability of the Megachiroptera. However, the index for their cerebellum is higher than in insectivorous bats; and this may again be related to their good climbing abilities.

Summarizing: The frugivorous Megachiroptera present forms which, considering the general development of their brain, occupy a very high position among Chiroptera (also WIRZ 1950). They are still further away than the frugivorous Microchiroptera from the marked and narrow specialization of the insect-eaters.

Fish-eaters (fig. 19e and table 5, column 5). This dietary type is represented here by one species only (*Noctilio leporinus*). The results therefore need confirmation and further study (e. g. of the Vespertilionid *Pizonyx*).

The neocortex and the cerebellum are somewhat larger than in the Megachiroptera. The cerebellum resembles, in its progression, that of the frugivorous Microchiroptera. The most original feature of the brain of this fish-eater is the high progression of the medulla oblongata and of the mesencephalon. These two structures are distinctly larger than in any other bat. Judging from what was found in other dietary types, this certainly relates functionally to the peculiar feeding habits of the animal. In the two sections of the brain concerned, the acoustic of the main sensory systems is well represented. In the Microchiroptera it is developed generally in the form of an ultrasonic device. To someone not acquainted with *Noctilio leporinus*, it may be difficult to understand how such a system can be utilized in catching a prey under water.

Observations by GRIFFIN (1958) and experimental studies reported by SUTHERS (1965) give clues how *Noctilio leporinus* may scan the water surface in flight and spot fish by echolocation. In parallel, our results confirm the marked importance of the acoustic system for that bat. Considering the size of the medulla oblongata and of the mesencephalon, the acoustic centers may even be larger in this species than in any other, in correspondence with the particular sonar differentiation required (penetration of water surface, refraction, adjustment for the lead-angle, etc.).

The olfactory system is reduced to the same extent as in the insectivorous species and obviously does not play any significant role in the special way of feeding which characterizes *Noctilio leporinus*. The structures of the limbic system (septum, hippocampus and schizocortex) are weak. This also agrees with what was found in the insecteaters. Possible functional relationships between the low development of these structures and some aspects of the behaviour will be discussed below in connection with the vampires.

Summarizing: In *Noctilio leporinus*, we have a form with a generally high development of the encephalon accompanied by very specialized feeding habits, in which the medulla oblongata and the mesencephalon play evidently an important role.

Blood suckers (fig. 19f and table 5, column 6). We were able to study only one representative species of the vampires, so that our results are to be regarded as provisional. The enlargement of the neocortex is more pronounced in *Desmodus* than in any other Chiroptera, including the Megachiroptera. The cerebellum is, in addition, the most developed among all the species so far investigated. This needs emphasis in view of the capability of these animals to move with great agility on the ground (like a large spider), to easily take to flight from the ground and to dexteriously crawl towards living prey: the vampire must approach cautiously, as for instance, the toes or ear-lobes of a sleeping man or the neck of a horse.

The olfactory centers are clearly less developed than in the frugivorous species, though they are better marked than in insect-eaters. From the present elementary analysis, one cannot say with certainty, which is the main sensory system of *Desmodus*. Well differentiated general acoustic structures (i. e. not predominantly ultrasonic) and the tactile system are probably important in this species.

Summarizing: Desmodus rotundus, with its very highly specialized feeding-habits (in fact, unique among mammals) appears to possess a high general development of the brain at the same time.

The small size of the limbic centers is quite remarkable in both blood and fish-eaters as already mentioned for *Desmodus* with regard to the hippocampus by SCHNEIDER (1957). The exact function of the limbic system is not known. It is certainly not specifically sensory and has accordingly been described by HASSLER (1964) as sense-independent. Among the behavioural changes following deficiency or experimental elimination of parts of that system, those which show increase of aggressivity or a loss of shyness (cit. in ANDY and STEPHAN 1964; HASSLER 1964) are of particular interest. It would be interesting to investigate by neuroanatomy to what extent the size of the limbic centers is related to the above differences in emotional behaviour. The present investigation concerning bats seems to indicate that such a relationship is indeed possible, in the sense, that species, which get their food by more aggressive methods, have smaller limbic structures. They all are bats which live on animal food (insect, fish, blood) whatever their evolutionary level (= neocortical development). It would seem especially promising to include in the study the carnivorous bats (*Vampyrum*, *Megaderma*), which were not available to us till now. In contrast, all frugivorous types (and the nectar-feeders as transitional forms) have clearly larger limbic structures. They are located together at the top of the progression scales for these structures.

Similar positions are observed on the scales for the olfactory centers and therefore one might be tempted to assume the existence of direct and narrow functional relationships between limbic and olfactory structures. However, experimental research and our size comparisons in Primates (STEPHAN 1966) make the assumption improbable.

We have attempted here to judge the peculiarities of the brain of Chiroptera in relationship to functional aspects. The attempt remains unsatisfactory because only elementary sensory and motory activities could be considered. When more elaborate behaviour such as aggressivity, indifference or fear is concerned it is possible to establish only vague relationships with distinct parts of the brain.

In fact, even for the elementary sensory and motory functions, the subdivisions of the brain so far established prove insufficient. Up to a point, the size of the cerebellum gives us a clue regarding the functional possibilities of the motory system, at least of one of its most important components. In comparison with the basal Insectivores, the cerebellum is very large in all bats. But in such relatively poor fliers as the Pteropids, it appears more progressive than in the nimble insect-hunters. We believe and we have already stressed that, on this point, the broadening of the locomotion spectrum seems to be of great significance. In the Pteropids, it is realized in the climbing habits of the animals searching for food in the trees; in *Desmodus* (which has the most progressive cerebellum of all species investigated) it expresses itself in the mobility on the ground and the careful, finely adjusted movements required to approach the skin of large blood-dispensing prey. That the forms with the largest cerebellum are not necessarily the swiftest and nimblest fliers (as, for instance, swallows) but species with a broader locomotion spectrum (such as some parrots, wood-peckers, divers or birds of prey) can be deduced also for birds from the work of PORTMANN (1947).

We can also obtain a fairly good idea of the functional capacity of the olfactory system from the size of its primary center (= olfactory bulb). According to our comparisons, olfaction is least important in the insect- and fish-eaters, and highest in the fruit-eating bats, especially in the Pteropids. This fits well with behavioural observations made by MÖHRES (1953) and others.

No definite information can be obtained by our measurements concerning the optic system. We have not yet distinguished the specific diencephalic and mesencephalic (or the neocortical) centers; we have only considered those parts of the brain in their totality. This is also valid for the acoustic system and for the dermal sensory systems, including gustation. More detailed investigation of finer structures is in progress.

In so far as comparable, the given results correspond clearly with former neuroanatomical and behavioural data given by various authors (e. g. concerning the different importance of olfaction in insect- and fruit-eaters). However, most previous authors consider the Microchiroptera as being uniform and simultaniously in contrast to the total of Megachiroptera. Our data have shown, that size and composition of the brain in Microchiroptera vary in a wide range. Many of the existing differences are obviously due to differences in life- and especially feeding-habits, and not to the affiliation of the various species to different systematic groups. This could be expected in regard to structures, which directly belong to a sensory or motory system. But it is also relevant to the highest sites of integration, the neocortical structures. This surprising result needs further discussion.

2. Modes of determination of the evolutionary level

We have utilized the neocorticalization as a criterion for judging the evolutionary level with the preliminary assumption, that this structure is the most progressive in Chiroptera. Since, however, this kind of assessment is not generally accepted in bats, it needs some justification. In literature the more generally used criterion is the more or less prominent flying ability of the members of this order. This mode can be followed back to WINGE (1892) and has been taken over without contradiction by SIMPSON (1945), GRASSÉ (1955), MANN (1963) and others. Facts about the brain were considered by MANN (1963) as of no importance in this respect and the attribution of a great significance to the neocortex as a yardstick of evolutionary progress, was discarded as an anthropocentric way of judgement. According to MANN, species with smaller neocortex should be more highly developed if they fly better. Correspondingly, the Megachiroptera were considered as the most primitive bats and within the Microchiroptera, the Rhinolophoidea to represent a primitive base, from which the other groups have derived and reached its highest forms in the Vespertilionoidea, especially in the Molossidae (WINGE).

Compared with our results, this mode of judgement leads to a quite different ascending sequence, with directly opposing positions of several groups. It is necessary, therefore, to establish, which of the two criteria is more significant in judging evolutionary progress: flying ability or neocorticalization. Certainly, both criteria are based on evolutionary processes, but both belong to quite different procedures of evolution. The flying ability represents a form of *specialization*, whereas the neocorticalization is an important expression of a process, which may be called "*integration*" or "*centralization*".

Our results in bats show, that the brain is affected by both processes, but obviously at different structural levels. In specialization, those structures of the brain become differentiated and enlarged, which belong to the functional systems, which are concerned with the specialization. These are primarily the lower centers of the brain stem up to the mesencephalon. In the course of centralization higher centers of coordination and association were built up, giving the animal a certain plasticity and versatility in its relationships to the environment. These centers include above all neocortex, striatum and part of the diencephalon. Certainly, the borders between these types of centers are not always sharp and not all parts of the brain can be linked to one of them.

In bats, the prominent flying ability of the insect-eaters has developed without stronger progression of the higher brain centers, whereas in Megachiroptera such centers exist coupled with a relatively modest flying ability. It therefore follows, that the two evolutionary processes occur more or less independently.

Recent groups are the result of branching. If a phylogenetic line plits, evolution can proceed along one branch with emphasis on further specialization without significant enlargement of the higher brain centers, and along another without further specialization but with an emphasis on development of those higher centers.

Examples to this effect are numerous, one of the most impressive is found in Primates. Within this order *Tarsius* is extremely specialized for jumping on prey in the dark, *Hylobates* to brachiation. *Homo*, though less specialized, has nevertheless the largest nerve centers and is generally regarded to be at the top of the evolutionary scale.

Nobody would regard *Tarsius* as the highest evolved primate, or if so, why not *Hylobates?* To compare the evolutionary level of the two species on the basis of specialization is impossible. Such modes of assessment have never been tried by serious scientists in orders with differently specialized groups. That it has been done for Chiroptera, is certainly due to their highly specialized locomotion, common in all bats. It gives the whole order its character and distinction from other mammals. Nevertheless, in this order also, the processes of evolution and the rules in evaluating evolutionary progress must be the same as for others.

We therefore believe, that specialization and its attendant symptoms cannot be used in evaluating evolutionary progress. Progressive evolution can best (or perhaps only) be judged and more evenly compared when based on the development of higher nerve centers. Our results in Chiroptera and the above examples in Primates are in full agreement with the concept of FRANZ (1935) that centralization is the precondition for progressive evolution[5]. One of the most relevant criteria in this respect is the size of the neocortex (WIRZ 1950).

3. Reflexions on the phylogenetic development of bats

Chiroptera are thought to originate from insectivore-like ancestors and therefore, we have based our comparisons on "basal Insectivores". To this group belong representatives of the tenrecs (Tenrecidae), hedgehogs (Erinaceidae) and shrews (Soricidae). The members of this group are less specialized and show the most primitive cerebral pattern of all placental mammals so far investigated. They are supposed to show the least degree of change since their first appearance ("permanent types"). Hence they can be expected to have remained comparatively close to the early forerunners of the Chiroptera, and therefore to represent a good reference basis for evaluating evolutionary progress.

We have shown, that the evolutionary level is best indicated by the neocortex as the brain structure most strongly increasing in size. The scale of increasing neocorticalization has been interpreted as an "ascending Chiroptera-scale". This scale represents of course no direct sequence of evolutionary stages in the sense, that the more progressive forms have passed through all the individual lower stages. The fish-eating *Noctilio* never has run through a fruit-eating *Artibeus*-stage. How then, can the ascending scale contribute to the problems of phylogenesis?

We believe, that the *development of higher nerve centers never reverts under normal conditions* (as distinct from the relationships in domestication and captivity). No

[5] According to him specialization develops without centralization.

opposing example is known to us. We can exclude from this assumption, that a form with a lower neocorticalization has derived from a form with a higher one. The size of the neocortex may therefore be used effectively in the discussion of phylogenetic sequences. Furthermore, it seems possible and likely, that the lowest recent forms under consideration correspond well with phylogenetic stages, through which higher forms have passed during phylogenesis.

a. Insectivora-Chiroptera transitional zone

On the basis of neocorticalization all insect-eating bats have remained most primitive, independent of their family affiliation. A strong overlap exists between the families and the variability within one may be larger than the differences between the mean values of various families. A well founded ascending sequence of families can not be performed within the insect-eating bats so far investigated. According to neocorticalization the insectivorous bats form a base stock, the forerunners of which certainly originated from terrestrial Insectivores and became specialized in the capture of flying insects presumably with intermediate arboreal and gliding stages before becoming capable of true active flight. As already mentioned, this specialization was not accompanied by an equally large increase in size of all parts of the central nervous system. Only those structures important for the specialization in question, have become comparatively large (cerebellum, mesencephalon, medulla oblongata).

There are nevertheless some species (*Asellia*, *Chaerophon*) in which even these structures have remained comparatively small. This is surprising in view of the fact, that the members of the Hipposideridae (including *Asellia*) and of the Molossidae (including *Chaerophon*) are very good insect-hunters. More comparative studies in the behaviour of these bats are certainly needed.

It is interesting in this connection to recall Möhres and Kulzer (1955) who have identified in *Asellia* an ultrasonic system which they describe as intermediary between the Vespertilionid and the Rhinolophid types. The authors speak of a higher level of functional complexity in *Asellia*. This appears to be in contradiction to the low size of medulla oblongata and mesencephalon, which is distinctly smaller than in both Vespertilionids and Rhinolophids, so far as investigated[6]. It may equally be possible that the genus represents a more generalized basal type from which further specialization developed. It would certainly be interesting to investigate the sonar system of *Chaerophon* as well, and to compare that of *Asellia* with terrestrial Insectivores in this respect.

In terrestrial Insectivores the ability for echolocation has been recorded so far in Soricidae and Tenrecidae (Gould et al. 1964; Gould 1965). Representatives of these families are included in our basal Insectivores. It is possible to assume, that echolocation already existed on the phylogenetic level of transition between Insectivora and Chiroptera, probably as a mere continuation of an Insectivora ancestor already endowed with such a system. We do not agree with Thenius (1969, p. 256), that Megachiroptera could have derived only from early Chiroptera, the echolocation of which was not yet developed in a manner characteristic for recent representatives. We believe, that the absence of echolocation in most Megachiroptera is due to a relatively less dramatic abandonment of the higher auditory frequencies. The existence of echolocation may even have been the most important prerequesite for the successful emergence of true Chiroptera. This system, which originally served mainly for orientation and/or social contact (as it presumably still does in terrestrial Insectivores today), then became increasingly important also for hunting prey in jumping/gliding/flying. Thus it has

[6] It is expected, that Baron's present studies of the size of the acoustic centers in medulla oblongata and mesencephalon will provide more detailed information on this problem.

finally developed, in all insect-hunting bats, as the specific sensory system to guide this function.

Differences in sound intensity related to feeding habits (high in insect and fish eaters, low in carnivorous and fruit eating bats) independent of family affiliation have been reported by GRIFFIN (1958, p. 250).

The centers of the olfactory system obviously underwent a strong reduction when bats originated. We assume here, on the basis of our knowledge of recent basal Insectivores, that the insectivore-like ancestors of the Chiroptera were macrosmatic forms. We believe that the reduction of the olfactory system occurred in approximate simultaneity with the development of flying ability, i. e. that it did not take place independently later. From the enlargement of the cerebellum and of the ultrasonic centers, one must anyway make that assumption, since these changes are by themselves a condition for the specialization concerned. With respect to the olfactory system, we base our assumptions on experience (especially with domesticated, captive and isolated animals) which indicated that the particular centers of a sensory system do not remain fully developed when the specific stimuli are partly or totally absent (STEPHAN 1950, HERRE 1966). A number of behavioural and neuro-experimental investigations have led to corresponding results (among others BENNETT et al. 1964; FIFKOVA and HASSLER 1969). Whether and when the reduction is subsequently fixed through heredity, is another question.

We have discussed here the probable simultaneity between development of flying ability and reduction of the olfactory structures because it seems to have a certain importance for the correct interpretation of fossil skulls (or skull fragments) and for their attribution to the order Chiroptera. We do not, for example, believe that the skull fragments described by EDINGER (1961) belong to a bat because the olfactory structures in it are too large. Posterior colliculi and olfactory bulbs of such large sizes as those illustrated and described by that author are found in terrestrial Insectivora (BAUCHOT and STEPHAN 1967) but not in bats, at any rate not in recent material. JEPSEN (1966) from another point of view also doubted, that the skull fragment in question is that of a bat.

b. Progressive Chiroptera

From the results of the present comparative neuroanatomical investigation, we expect the insectivorous bats to be the oldest phylogenetically. From this basal type, all others are thought to have derived. As far as the frugivorous species are concerned, it comes readily to mind that they may have developed through forms which hunted for insects on flowers and so became nectar-feeders and finally fruit-eaters[7]. In many characters of their brains, the nectar-feeders are an intermediate stage between insect-eaters and frugivorous bats (see fig. 19). The adaptation to a frugivorous diet, which implies a significant widening of the living space of bats, was accompanied not only by an increase of the higher brain centers, but also by an enlargement of the olfactory structures, a dubious fact at first sight. Starting from basal Insectivores, the proposed evolu-

[7] We give credit to this hypothetical phylogenetic sequence, though in recent nectar-feeders snout and tongue are longer and more slender than in both insect- and fruit-eaters. These characteristics are, however, at least in part due to specialization, and (as we have shown in the case of flying-ability) not useful to evaluate evolutionary progress and phylogenetic sequences. Two interpretations are possible (1) snout and tongue of the true nectarivorous intermediate stages were short and became longer only later in phylogeny or (2) they were long in the intermediate stages and shortened later in adaption to a frugivorous diet.

EISENTRAUT (1945) believes that the nectar-feeders among the Megachiroptera developed through fruit-eating stages. The relationships in the Phyllostomatidae indicate differently. But our material of Megachiroptera is too restricted, to enable us to draw a final conclusion.

tionary process would indeed imply that the centers of the olfactory system after a strong reduction in the transition to insect-eating bats, have enlarged again during the adaptation to a vegetarian diet and finally reached the original size (i. e. that of the basal Insectivores) as demonstrated in the Megachiroptera. This course of evolution is much more probable than a direct passage from primitive (basal) Insectivores to frugivorous bats, which, to us, appears difficult to perceive. The partial reduction of a sensory system in the course of evolution followed by re-enlargement, can be easily understood. The sensory centers, by nature, depend more heavily on environmental influences and on feeding-habits than the higher brain centers.

Fish-eating and blood-sucking bats certainly did not pass through nectarivorous and frugivorous stages in the course of their evolution. The blood-suckers may have evolved through carnivorous stages. Unfortunately, we were not able to investigate any true carnivorous species (*Vampyrops*, *Megaderma*). The fish-eaters may have evolved directly from insectivorous species. In favour of this view, we have among other arguments, the narrow relationships that still exist between fish- and insect-eaters in the Noctilionids, where one finds both diets in two species of the same genus[8].

Although we were so far unable, to study the brain structures of the insectivorous *Noctilio labialis*, the species seems to be, on the basis of total brain size, much more primitive than its piscivorous relative *Noctilio leporinus* (Pirlot and Stephan 1970).

It is worth emphasizing that not all bat families present a diversity of feeding-habits: within the Microchiroptera, this is only encountered to a significant extent in the New World Phyllostomatids. This single family includes nectar-feeders, frugivorous and carnivorous species as well as true insect-eaters. It would be of interest to investigate whether the insect-eaters are also the most primitive representatives in that otherwise highly evolved family. Our study of the encephalization points to such a conclusion (Pirlot and Stephan 1970). We expect that our projets for more extensive research on that peculiar family will enable us to confirm the evolutionary processes hypothesized and suggested above.

The interesting problem concerning the more specific origin of the Old World frugivorous Megachiroptera must also remain open. Did the Old World bats have in phylogeny such a strongly diversified "phyllostomatoid" family? We do not have sufficient data from our neuro-anatomical researches to discuss this question.

But, on the basis of neocorticalization the question concerning the general evolutionary level of Megachiroptera can be answered: it is high. Our results confirm the results of Wirz (1950) and Mangold-Wirz (1966) and are in agreement with the view of Griffin (1958), Thenius and Hofer (1960) and Thenius (1969), that the Megachiroptera are geologically younger than the Microchiroptera. Their absence in South America also argues in favor of this conception (Baker and Harris 1957; Thenius 1969). Some contrary conceptions based on their flying ability have already been discussed. In addition, Schneider (1957) supports an opposing view on some brain criteria (length of the hemispheres, position of the corpus callosum and development of the optic system). According to Schneider, the brain of the Megachiroptera, rather than that of the Microchiroptera, corresponds to the basal type of the Insectivores. However, the criteria used are of restricted importance in judging evolutionary progress and the cortical surface measurements made by Schneider confirm our own results, when based on a comparison of the neocortical surface with regard to body size.

[8] Some authors have made them two genera, *Noctilio* and *Dirias*. In any case, these animals closely resemble one another, differing mainly in general size and in the length of the hindfoot.

4. Evolutionary level, systematic relationship and dietary type

The results of this quantitative analysis show, that there exists no close affiliation between evolutionary level and systematic relationship, as already indicated by our research on encephalization (PIRLOT and STEPHAN 1970). The evolutionary level can be similar within different families (Rhinolophids, Hipposiderids, Vespertilionids, Molossids) and quite divergent within one family (Phyllostomatids) or even genus (*Noctilio*). The degree of encephalization or neocorticalization cannot therefore be used in evaluating the systematic relationships of bats. This fact is, however, not a specific characteristic of this order, but a common fact, generally in existence in widely ramified groups. A peculiarity of bats nevertheless exists in the close relationships between the various dietary types and definite evolutionary levels, an interdependence so far not observed in other groups, for instance in Primates (of which we have so far the most extensive information on brain measurements). On the ascending primate scale, the carnivorous, vegetarian and omnivorous types appear in a mixed succession. An explanation of the peculiarities of bats is difficult and we wish to stress, that the following interpretations can only be hypothetical.

We believe, that the close interdependence between dietary type and evolutionary level in bats must be referred to the unusually strong specialization of the basic types. The early evolution of bats was obviously directed towards the achievement of the flying-ability and the improvement of the preying ability by means of the ultra-sonic device combined with a simultaneous maintenance of a minimal body weight. Originating from such a basic group, every further evolution could develop only through a transitional de-specialization with respect to new types of food correlated with a de-specialization in locomotion and sensation. The stage of broadening the feeding and/or locomotion and sensation spectrum must have existed even when another restriction took finally place as, for example, in the case of the blood-suckers. Regarding the dietary type, the evolution of the vampires proceeded presumably via an intermediate insectivorous/carnivorous stage, and that of the pure fruit-eaters via an insectivorous/nectarivorous/frugivorous succession. The locomotion ability for hunting insects was broadened by hovering for the capture of insects on flowers and by the improvement of mobility on the ground and in branches. In relation with this widening of the feeding and/or locomotion spectrum, the representation of these functions in the central nervous system became reinforced, the unilateral dominance of a single sensory system[9] reduced and the brain more differentiated as a result. The development of the higher association and coordination centers are no doubt also interrelated with that process; such development having possibly been prevented in the basic types by the exceptionally marked dominance of a single sensory system, i. e. the acoustic.

The most universal types, as regards the development of the sense organs, are the fruit-eaters. Their acoustic system, as well as their optic and olfactory structures are well developed, so is presumably their gustatory system. An investigation of the latter is now in progress (PIRLOT). On the other hand, a certain regression of the flying-agility has taken place in the fruit-eaters. They have retained the flying-ability but with less perfection; and have abandoned the habits of capturing insects on the wing, which had originally led to flying and adapted to feeding on immobile fruits. Flying-ability though no longer strictly necessary, evidently remains advantageous since it renders protection and enables the animals to reach feeding haunts more rapidly and in a wider area. If flying-ability would have resulted in a handicap to a group of bats, non-flying bats would no doubt exist.

[9] NEUWEILER (1966) also emphasizes the predominance of echolocation in Microchiroptera and characterizes the other sensory systems to have only auxiliary functions.

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Summary

1. The volume size of 11 components of the brain has been compared within Chiroptera (18 species from 8 families) and also with those of primitive "basal" Insectivores by aid of the allometry formula.

2. Progression indices give direct numerical information as to how many times a structure has enlarged in comparison to its equivalent from a basal Insectivore of equal body weight.

3. The relationship between the degree of progression of each brain component and life-habits are stressed and discussed, especially for the cerebellum in relation to flying abilities and for the mesencephalon and medulla oblongata in relation to ultra-sonic echolocation. Limbic structures are better developed in vegetarien types than in bats with an animal diet. The olfactory bulb shows a clear although varied decrease in all groups (strongest in insect-eaters) except in the frugivorous Megachiroptera.

4. The great variability of certain structures (especially of those linked to specific functions) and the independence of their development from that of the neocortex are emphasized.

5. The structure with the strongest enlargement is the neocortex. Its size is therefore used to evaluate the evolutionary level reached by each species. The scale of increasing neocorticalization is interpreted as an "ascending Chiroptera scale".

6. The position in the ascending scale cannot be used to evaluate systematical relationships. The evolutionary level can be divergent within one family and similar in different families.

7. The existence of a clear relationship between type of feeding habit and neocorticalization is indicated, as was already found for encephalization studied in a more extensive material (Pirlot and Stephan 1970). Within the ascending Chiroptera scale all insect hunting bats (so far investigated) have by far the lowest positions ("basal Chiroptera", with Rhinolophids, Hipposiderids, Vespertilionids and Molossids as representatives) followed by the nectarivorous and frugivorous Phyllostomatids. The slower and heavy flying Megachiroptera take a high position and are surpassed only by the fish-eating *Noctilio leporinus* and the blood-sucking *Desmodus rotundus*. Similar ascending trends are found also in striatum and diencephalon.

8. The ascending Chiroptera scale represents of course no direct sequence of evolutionary stages in the sense, that the more progressive forms have passed through all the individual lower stages. However it seems possible and likely, that the lowest recent forms under consideration correspond well with phylogenetic stages, through which higher forms have passed during phylogenesis. Furthermore it is improbable that the development of higher nerve centers reverts under wild life conditions (in contrast to domestication and captivity). The size of the neocortex is therefore considered an effective guide in the discussion of phylogenetic sequences.

9. If these assumptions (paragraph 8) are correct the results of our analysis can be interpreted as follow:

a. the phylogenetic oldest bats have been insect-eaters. They may have evolved from primitive Insectivores essentially through specialization for catching insects on the wing by means of the ultra-sonic device. They form a basis group from which all other bats derived.

b. Each progressive evolution of the brain is related to a broadening of the feeding and/or locomotion spectrum and has led to nectar feeders and frugivorous types on the one hand, carnivorous, fish-eating and blood-sucking types on the other.

c. The unilateral dominance of a single sensory system (echolocation) becomes reduced, a process which can be regarded as "de-specialization". The progressive evolution of higher brain centers, still very small in basal Chiroptera, is thought to be related to that process.

10. The above interpretation of the results of our measurements is in contradiction with the predominant conception about probably existing phylogenetic sequences in bats. This is due to the unilateral evaluation of the evolutionary process of specialization (improvement of flying ability) under simultaneously existing neglect of the evolutionary process of centralization (development of higher structures in CNS). The two processes develop almost independently from one another, as can be shown clearly in bats. The existence of centralization or integration (not specialization) is the precondition of progressive evolution and its assessment.

Zusammenfassung

1. Die Volumengrößen von 11 Hirnstrukturen bei 18 Fledermausarten aus 8 Familien wurden unter Verwendung der Allometrieformel untereinander und mit jenen primitiver „basaler" Insektivoren verglichen.

2. In Form von Progressionsindices wird direkt in Zahlen angegeben, um wievielmal sich eine Struktur im Vergleich zu jener eines basalen Insektenfressers gleichen Körpergewichts vergrößert hat.

3. Die Beziehungen zwischen dem Ausmaß der Vergrößerung und der Lebensweise werden erörtert, insbesondere für das Kleinhirn in Beziehung zur Flugfähigkeit und für Mittelhirn und Medulla oblongata in Beziehung zum Ultraschallvermögen. Die limbischen Strukturen sind bei den Fledermäusen mit vegetarischer Nahrung besser entwickelt als bei jenen mit tierischer Nahrung. Der Bulbus olfactorius zeigt bei allen Gruppen, mit Ausnahme der fruchtfressenden Megachiropteren, eine deutliche Abnahme (am stärksten bei den insekten-jagenden Chiropteren).

4. Die starken Unterschiede in der Progression einiger Strukturen (insbesondere solcher, die mit spezifischen Funktionen in Verbindung gebracht werden können) und ihre Größenunabhängigkeit vom Neocortex werden hervorgehoben.

5. Die sich am stärksten vergrößernde Struktur ist der Neocortex. Seine Zunahme verwenden wir deswegen als Maßstab für die Beurteilung des erreichten Evolutionsniveaus; die Skala der Größenzunahme wird als „Aufsteigende Chiropteren-Reihe" interpretiert.

6. Die Position in der aufsteigenden Chiropterenreihe kann nicht als Maßstab für den Verwandtschaftsgrad zwischen den Arten gewertet werden. Innerhalb einer Familie kommen Arten unterschiedlichen Niveaus vor, während verschiedene Familien gleiches Niveau haben können.

7. Wie bei den an breiterem Material durchgeführten Untersuchungen zur Encephalisation (= Vergleich des Gesamtgehirns; PIRLOT und STEPHAN 1970) ergeben sich deutliche Beziehungen zum Ernährungstypus. In der aufsteigenden Chiropterenreihe (= Neocorticalisation) haben alle bisher untersuchten insekten-jagenden Fledermäuse die deutlich niedrigsten Positionen („Basale" Chiropteren mit den bisher untersuchten Familien: Rhinolophidae, Hipposideridae, Vespertilionidae, Molossidae), gefolgt von den nektar- und fruchtfressenden Phyllostomatiden. Die langsamen und schwerfälliger fliegenden Megachiropteren haben eine hohe Position und werden nur vom fischfressenden *Noctilio leporinus* und vom blutleckenden *Desmodus rotundus* übertroffen. Ähnliche ansteigende Trends wie beim Neocortex finden sich auch bei Striatum und Diencephalon.

8. Die aufsteigende Chiropteren-Reihe stellt keine direkte Folge phylogenetischer Stadien in dem Sinne dar, daß die mehr progressiven Formen alle niedrigeren Stadien der Reihe nach durchlaufen haben. Es ist jedoch möglich und wahrscheinlich, daß die niedersten rezenten Formen in den hier untersuchten Merkmalen solchen phylogenetischen Stadien ähneln, die von höheren Formen während deren Phylogenese durchlaufen wurden. Rückbildung des Neocortex während der phylogenetischen Entwicklung ist unwahrscheinlich, so daß dessen Größe bei der Diskussion phylogenetischer Sequenzen bevorzugt herangezogen werden kann.

9. Unter diesen Voraussetzungen (Punkt 8) lassen sich die Ergebnisse wie folgt interpretieren:

a. die phylogenetisch ältesten Fledermäuse waren insektenfressende Formen. Sie sind wahrscheinlich aus terrestrischen Insektivoren durch Spezialisation auf den Fang fliegender Insekten mit Hilfe von Ultraschall entstanden. Sie bilden eine Basisgruppe, aus der alle anderen Fledermäuse hervorgingen.

b. Jegliche progressive Hirnevolution steht bei den Fledermäusen in Beziehung zu einer Erweiterung des Nahrungs- und/oder Bewegungsspektrums und hat zu Nektar- und Fruchtfressern einerseits, zu fleischfressenden, fischfressenden und blutleckenden Typen andererseits geführt.

c. Die einseitige Dominanz eines einzelnen sensorischen Systems (Ultraschall) wird reduziert, ein Prozeß, der als „Entspezialisierung" betrachtet werden kann. Die progressive Entwicklung höherer Integrationszentren, die bei den basalen Chiropteren noch sehr klein sind, könnte damit im Zusammenhang stehen.

10. Die vorstehende Interpretation der Ergebnisse unserer Messungen steht in Widerspruch zu den vorherrschenden Vorstellungen über mögliche phylogenetische Sequenzen bei den Fledermäusen. Die Ursache liegt in der bisher üblichen einseitigen Bewertung des evolutionären Prozesses der Spezialisation (Vervollkommnung des Flugvermögens) bei Vernachlässigung des evolutionären Prozesses der Zentralisation (Entwicklung höherer zentralnervöser Strukturen). Die Prozesse verlaufen weitgehend unabhängig voneinander, wie gerade die Verhältnisse bei den Fledermäusen zeigen. Voraussetzung für eine *progressive* Evolution und für die Möglichkeit einer vergleichenden Beurteilung des Evolutionsniveaus ist die Zentralisation oder Integration, nicht die Spezialisation.

Résumé

1. Les volumes de 11 structures encéphaliques ont été mesurés chez 18 espèces de Chiroptères appartenant à 8 familles; les valeurs ainsi obtenues ont été comparées entre elles et avec celles des Insectivores de base primitifs, en utilisant la méthode de régression allométrique.

2. Les résultats sont donnés directement sous forme d'indices de progression, qui indiquent combien de fois une structure a augmenté de volume par rapport à celle d'un Insectivore de base de même poids somatique.

3. L'importance de l'augmentation des centres nerveux a été rapprochée de certains aspects de la biologie de ces animaux, en particulier le cervelet de l'aptitude au vol, ou le mésencéphale et la moëlle allongée de l'écholocation. Les structures limbiques sont plus développées chez les Chauves-Souris à régime herbivore que chez celles qui sont carnivores on sens large. Le bulbe olfactif montre dans tous les groupes, à l'exception des Mégachiroptères frugivores, une réduction nette (très accusée chez les Chiroptères entomophages).

4. On peut mettre en évidence le contraste net qui existe entre les indices de progression de quelques structures encéphaliques (en particulier celles qui sont liées à des fonctions spécifiques), et leur indépendance vis-à-vis de la progression néocorticale.

5. La structure encéphalique qui augmente le plus en volume est le néocortex. C'est pourquoi nous l'utilisons comme base de référence pour estimer le niveau évolutif atteint par chaque espèce; l'échelle ainsi obtenue est considérée comme une «échelle ascendante des Chiroptères».

6. La position des diverses espèces dans cette «échelle ascendante» ne peut être utilisée comme critère de parenté. Au sein d'une même famille, les espèces peuvent se situer à des niveaux très variés, tandis que des familles distinctes peuvent se situer à des niveaux comparables.

7. Comme pour le matériel plus abondant que nous avons analysé dans l'étude de l'encéphalisation (Pirlot et Stephan 1970), on peut mettre en évidence ici des relations nettes avec le régime alimentaire. Dans l'échelle ascendante de néocorticalisation les Chauves-Souris entomophages sont en position nettement inférieure (ce sont les «Chiroptères de base», comprenant, parmi les familles étudiés jusqu'à présent, les Rhinolophidae, les Hipposideridae, les Vespertilionidae et les Molossidae), suivies des Phyllostomatidae nectarivores et frugivores. Les Mégachiroptères au vol lent et lourd ont une position élevée et ne sont dépassées que par *Noctilio leporinus*, qui est piscivore, et par *Desmodus rotundus*, qui est sanguinivore. Des tendances analogues à celles qu'indique le néocortex sont fourniers par les indices correspondant au striatum et au diencéphale.

8. Cette échelle ascendante des Chiroptères ne prétend évidemment pas représenter une succession phylétique directe, et l'on ne peut affirmer que les espéces les plus évoluées sont passées par tous les stades qui les précèdent dans la série. Il est néanmoins possible, voire probable, que les espèces actuelles les plus bas situées ressemblent dans les structures que nous étudions aux stades qui ont été parcourus, au cours de leur phylogenèse, par les espèces haut situées. Comme la régression du néocortex au cours de la phylogenèse est hautement improbable, on utilisera de préférence les indices fournis par cette structure pour la discussion des sequences phylogénétiques.

9. Dans les conditions qui viennent d'être définies au point 8., voici les résultats de notre étude:

a. les plus anciennes Chauvis-Souris dans la phylogenèse etaient entomophages; elles proviennent vraisemblablement d'Insectivores terrestres qui se sont spécialisés dans la capture d'Insectes en vol grâce à l'écholocation. Elles forment un groupe basal, d'où proviennent tous les autres Chiroptères.

b. chaque étape de l'évolution encéphalique est liée chez les Chauves-Souris à un ênlargissement du spectre des modes d'alimentation et/ou des modes de locomotion; cette progression a conduit d'une part vers les espèces nectarivores et frugivores, d'autre part vers les espèces carnivores, piscivores et sanguinivores.

c. l'influence dominante d'un système sensoriel unique, l'écholocation, a donc régressé, et l'on peut qualifier un tel processus de «déspécialisation». Le développement progressif de centres nerveux supérieurs, qui sont encore de faible taille chez les Chiroptères de base, peut être mis en parallèle avec ce processus.

10. L'interprétation que nous donnons aux résultats de nos mesures vient en contradiction avec les vues les plus courantes sur la phylogénie des Chiroptères. La raison réside dans le fait qu'on n'avait pris en considération, jusqu'ici, que le seul processus évolutif de la spécialisation (perfectionnement des capacités de vol), et négligé le processus évolutif de la centralisation (développement des centres nerveux supérieurs). Ces deux processus se produisent en fait indépendamment l'un de l'autre, comme le montrent précisément les Chiroptères. C'est la centralisation ou intégration qui traduit l'évolution progressive et qui permet une estimation du niveau évolutif atteint par les diverses espèces.

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BUCHBESPRECHUNGEN

Geus, Armin: **Sporentierchen, Sporozoa. Die Gregarinida der land- und süßwasserbewohnenden Arthropoden Mitteleuropas.** In: Die Tierwelt Deutschlands, 57. Teil. Jena: VEB Gustav Fischer 1969. 608 S., 338 Abb. Brosch. 131,30 M.

Eine auf den neuesten Stand der Systematik gebrachte Monographie der Eugregarinenfauna der Land- und Süßwasserarthropoden Mitteleuropas mit übersichtlichen Bestimmungstabellen und durchwegs nach dem Leben entworfenen Strichzeichnungen. Unter den berücksichtigten 322 Arten mußten 2 Gattungen: *Neomonocystis* und *Torogregarina* sowie 75 Spezies als neu erstmalig beschrieben werden. Im allgemeinen Teil werden die Morphologie, Feinbau und Bewegungsmechanik, der Entwicklungsablauf, Geschlechtsbestimmung und Sexualdimorphismus, die Pathogenität, das Parasit-Wirtsverhältnis, die Wirtsspezifität und die Untersuchungstechnik behandelt. Der weit umfangreichere spezielle Teil enthält die Bestimmungsschlüssel und die Beschreibung der einzelnen Formen, nach Familien und Gattungen geordnet. In seiner Vollständigkeit vorbildlich ist das jeder Art beigegebene Verzeichnis der Synonyma. Ein umfangreiches Verzeichnis der Wirtstiere beschließt den speziellen Teil, der für alle diejenigen, die in Zukunft über Eugregarinen arbeiten wollen, eine unentbehrliche Grundlage darstellt. Die Originalzeichnungen und Größentabellen leisten dabei gute Dienste.

Die Gregarinen umfassen bekanntlich die sich ausschließlich gamontogam fortpflanzenden Eugregarinen und die durch einen Wechsel von Schizogonie mit Gamogonie charakterisierten Schizogregarinen. Letztere werden, obwohl zu den häufigeren Arthropodenparasiten zählend, in der vorliegenden Monographie weggelassen, was zweckmäßigerweise bereits im Titel der Arbeit hätte angegeben werden sollen. Wünschenswert wäre die Wiedergabe des Schemas eines typischen Entwicklungsablaufes, etwa von *Stylocephalus*, im allgemeinen Teil zwecks Veranschaulichung des Kernphasenwechsels und der sonstigen Vermehrungsvorgänge. Die für die Beurteilung der systematischen Stellung der Gregarinen so wichtige Parallele: Eugregarinida-Eucoccidia, Schizogregarinida-Schizococcidia bleibt unberücksichtigt.

Imponierend ist die Arbeitsleistung des Verf. und seiner Mitarbeiter bei der Bewältigung der zahlenmäßig so umfangreichen, durch ihre Einförmigkeit wenig attraktiven, jedoch biologisch hochinteressanten Protozoengruppe.

E. Reisinger